



FORD - 9141

FORD-9141 PROTOCOL DEFINITION

AND

INTERFACE REQUIREMENTS

CPSC #00.06

Vehicle Communications Link

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AVT Electrical/Electronic Systems Engineering

Body E/E Modules Subsystems Department

Multiplex Standards and Applications Section



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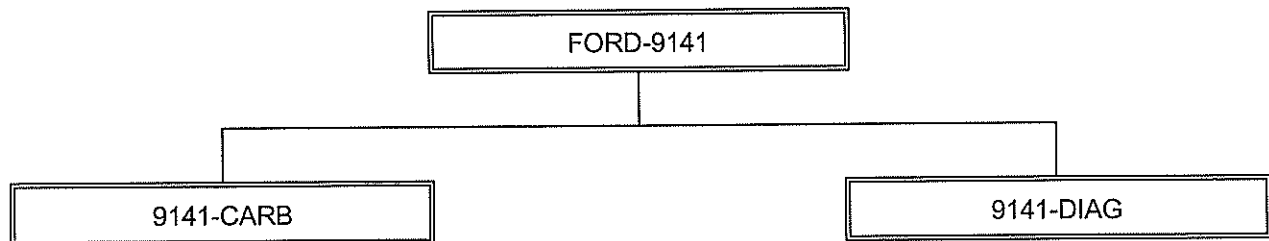
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1.0 GENERAL

This specification describes a protocol suitable for use in a vehicle diagnostics application. The diagnostics application provides potential improvement in the manufacture and service of vehicle electrical systems. Using the capabilities provided in this methodology, on-board ECU's shall communicate with on-board testers, in-process and end-of-line monitors, and service bay testers and diagnostic systems. Information transfer techniques are used to accomplish control, monitor, interrogation, substitution, and calibration functions.

This methodology is based upon the ISO-9141 standard and shall be designated the FORD-9141 Diagnostic System. The FORD-9141 implementation is comprised of two segments. The first segment addresses compliance with California Air Resources Board (CARB) requirements. It is referred to as 9141-CARB. The Ford implementation of 9141-CARB is defined by, and is totally compatible with, SAE J1979 and ISO 9141-CARB. The other segment addresses the internal needs of Ford Motor Company to manufacture and repair vehicles. This segment is referred to as 9141-DIAG. The two methodologies are totally compatible at the hardware level and may share common link hardware.





1.1 PURPOSE

This specification defines the FORD-9141 communication protocol. It is intended to be equally applicable to both on-board modules and off-board diagnostic testers.

9141-CARB will be implemented for the purpose of CARB compliance. It is fully compliant with the implementation described in the ISO-9141-CARB specification and SAE J1979.

The 9141-DIAG implementation will be used by the service community to diagnose and repair vehicles. It is supported in the New Generation Star (NGS) Tester, the Low Line Tester (LLT), and the Service Bay Diagnostic System (SBDS).

9141-DIAG may be used in the manufacturing environment to verify the proper assembly of vehicles in-process and at end-of-line. NGS units and specially designed units may be used to verify the integrity of harness connections and the presence of door switches, sensors, lamps, heating elements, relays, and motors.

Module integrity will be assured using 9141-DIAG to verify functionality prior to shipment for assembly. Computer driven test stands will execute test programs via the link. Executable code may be downloaded to the module, executed, and the results uploaded. In addition, 9141-DIAG may be used to transfer application specific calibrations and configuration information to the module.

1.2 SCOPE

This specification and DS-YF1T-1A294-AB - "Global Diagnostic Specification", are the controlling documents that determine if a diagnostic communication system or network has achieved compatibility with the FORD-9141 protocol definition. It defines the FORD-9141 communication methodology applicable to diagnosis or manufacture a vehicle. This specification does not address the communication methodology required for inter-module transfer of information during normal "Down-the-Road" vehicle operation.

The document specifies the parameters necessary for proper implementation of the bus.



1.3 USE

The 9141-CARB methodology qualifies as a bus communication protocol that conform to the requirements of the California Air Resources Board for combustion by-product emissions.

The 9141-CARB methodology may be applicable to a low line vehicle powertrain application.

The 9141-DIAG methodology is suitable for applications that incorporate microprocessors that do not require communication with other modules on the vehicle. In these cases, 9141-DIAG is a cost effective solution. In applications that require a network for intermodule communication, 9141-DIAG should not be used because it does not support bus arbitration and contention resolution.

1.4 DESCRIPTION

This serial communication link is used for diagnostic communication. The communication rate is fixed at a nominal rate of 10.4 Kbits/second (the exact frequency is derived from 1 bit per 96 msec) for all systems. Information transfer is fully bidirectional (any device can transmit or receive). The information transfer method uses messages to communicate via a network or a simple two party link. Transactions follow a master/slave procedure. Allocation of the bus shall be controlled, because the bus contains only one communication path and no contention resolution exists. Consequently, the master shall initiate all transactions. Slaves shall never initiate transactions. The master is responsible for apportioning possession of the bus and preventing the occurrence of collisions.

In all cases, the diagnosed ECU is the slave and the diagnosing unit is the master. For example, in the service bay environment the SBDS will play master to any on-board module which performs as slave. In the case when the diagnosing unit is an on-board instrument cluster, the cluster shall function as master to the diagnosed slave ECU.



2.0 APPLICABLE DOCUMENTS

The following documents are either referenced by this document or are related to this document. Unless otherwise specified, if a conflict between this document and other applicable documents exists, the following order of precedence shall apply:

- This Document
- Specific Ford Purchase Order / Release Document
- Specific Ford Module Engineering Specification (ES)
- Applicable Documents referenced herein

2.1 FORD SPECIFICATIONS

ES-F10F-1316-AA Specification - Electromagnetic Compatibility

DM17.01-351 section 4.4.3 - "Low Voltage Wiring" Corporate Engineering Design Standard

DS-YF1T-1A294-AB - "Global Diagnostic Specification"

2.2 SAE and ISO DOCUMENTS

J1213 Glossary of Vehicle Networks for Multiplexing and Data Communications

J1850 Class B Data Communication Network Interface

J1962 CARB Standard Diagnostic Connector

J1978 OBDII Scan Tool

J1979 E/E Diagnostic Test Modes

J2190 Enhanced Diagnostic Test Modes

J2012 Diagnostic Trouble (Fault) Codes

ISO-9141 Requirements for Interchange of Digital Information



3.0 DEFINITIONS / ACRONYMS

The definitions provided in the SAE J1213 Glossary apply to this specification. Additional or modified definitions of specific terms unique to this document are listed below:

BPS	-	Bits per second.
Byte	-	A group of eight bits of data transmitted along the serial link beginning with the least significant bit.
CARB	-	California Air Resources Board (State Agency).
DMR	-	Direct Memory Read. A method of obtaining information from an ECU by accessing its memory address space.
ECU	-	Electronic Control Unit. An onboard electronic module that controls a vehicle subsystem function.
EPA	-	Environmental Protection Agency (Federal Agency).
h	-	suffix denoting a hexadecimal number.
ISO	-	International Standards Organization
Nibble	-	Four bits of data. A byte may be split into two nibbles.
NGS	-	New Generation Star Tester
OBDII	-	On-Board Diagnostics II. Set of emission related stipulations that are being required by the California Air Resources Board.
Offset	-	a number used to refer to a data value by specifying its relative position in a list of data values.
PID	-	Parameter Identifier. An index number used to refer to a parameter within a module without knowing its storage location.
SAE	-	Society of Automotive Engineers
SBDS	-	Service Bay Diagnostic System
Tester	-	Any offboard device that may cause an ECU to enter a test mode or communicate stored codes, PIDs, or status.
V _B	-	Vehicle Battery Voltage



4.0 NETWORK STRUCTURE

Connection is accomplished using wire connections between communicating ECU's. The interconnection shall couple as few as two ECU's (including a tester) and as many as 20 onboard ECU's on a single bus or network. The wire may be arranged in a bus configuration and routed throughout the vehicle, or a separate wire may be provided to connect the testing unit to each ECU under test.

4.1 CONFIGURATION

A single signal wire shall be used to communicate bidirectionally. This corresponds to ISO 9141 configuration #5 (bidirectional K line without inclusion of L line). All units on the communication bus shall be capable of transmitting and receiving on this wire.

No initialization is required to begin communication. All Ford-9141 communication protocols are common and use a single fixed baud rate.

Initialization for 9141-CARB is described in section 6.1 of this document.

4.2 DATA TRANSMISSION RATE

Information shall be transmitted at a rate equal to 1 bit per 96 msec (or nominally 10.4 Kbits/second). All Tester units shall transmit at the given rate $\pm 1\%$, and shall correctly receive transmissions that deviate less than $\pm 3\%$. All ECU's shall transmit at the given rate $\pm 2\%$, and shall correctly receive transmissions that deviate less than $\pm 2\%$.

All ECU's shall be capable of maintaining full module functionality while responding to network interrupts at the maximum rate. All ECU's shall be ready to receive 100% message traffic once the K-line is pulled to logic "1" for 50 milliseconds. In the event that the bus is filled to 100% capacity by valid message traffic or noise due to a termination or wiring fault, the ECU may be interrupted at one-tenth of the baud rate. For the FORD-9141 protocol, ECU's may be interrupted with irrelevant information as frequently as every 960 microseconds at 100% bus utilization. The ECU shall not ignore the interrupt since relevant bytes may arrive at any time.

4.3 SIGNAL

Information on the bus is transmitted in the Non-Return to Zero (NRZ) encoding format. In NRZ encoding, each bit is represented by a high or low level for the entire duration of the bit time. No clock timing is embedded in the bus signal as is the case with pulse width modulated encoding. Timing is referenced to the first edge of the start bit of each message.

4.3.1 TRANSMITTED LEVELS

All signal level requirements are referenced to the vehicle battery (V_B). Maximum ground offset voltages must be accommodated when designing to meet transmission and reception level requirements.

In addition, the transition time of all bits (measured at the transmitter) shall be less than 9.6 microseconds (10% of 1 bit time). The transition time is measured as the time from logic "1" to logic "0" or the time from logic "0" to logic "1". This is defined as the time from the 20% V_B point to the 80% V_B point or the time from the 80% V_B point to the 20% V_B point.



4.3.1.1 ECU TRANSMITTED LEVELS

ECU transmitters shall generate a voltage greater than 80% of V_B to create a valid logic "1".

ECU transmitters shall generate a voltage less than 20% of V_B to create a valid logic "0".

The table below indicates the WORST CASE transmitter driven bus levels for 8, 12, and 16 volts:

TRANSMITTED LEVEL	$V_B = 8$ volts	$V_B = 12$ volts	$V_B = 16$ volts
min Logic "1" (80%)	> 6.4 volts	> 9.6 volts	> 12.8 volts
max Logic "0" (20%)	< 1.6 volts	< 2.4 volts	< 3.2 volts

4.3.1.2 TESTER TRANSMITTED LEVELS

Testers shall generate a voltage greater than 90% of V_B to create a valid logic "1".

Testers shall generate a voltage less than 10% of V_B to create a valid logic "0".

The table below indicates the WORST CASE tester driven bus levels for 8, 12, and 16 volts:

TRANSMITTED LEVEL	$V_B = 8$ volts	$V_B = 12$ volts	$V_B = 16$ volts
min Logic "1" (90%)	> 7.2 volts	> 10.8 volts	> 14.4 volts
max Logic "0" (10%)	< 0.8 volts	< 1.2 volts	< 1.6 volts

4.3.2 RECEIVER DETECTION

To ensure proper operation, implementation shall afford at least 1 volt of detection margin to reject coupled noise and voltage offset when the signal is at logic "0" or logic "1".

A voltage level of more than 70% of battery voltage (V_B) at the receiver shall be detected as a logic "1". The threshold (V_{th1}) for detection of transition from logic "0" to logic "1" shall be set at any value in the range 30% V_B to 70% V_B which is higher than the threshold voltage V_{th0} . A voltage level of less than 30% of battery voltage (V_B) at the receiver shall be detected as a logic "0". The threshold (V_{th0}) for detection of transition from logic "1" to logic "0" shall be set at any level in the range 30% V_B to 70% V_B , which is lower than the threshold voltage V_{th1} .

The table below indicates required detection thresholds for 8, 12, and 16 volts:

THRESHOLDS	$V_B = 8$ volts	$V_B = 12$ volts	$V_B = 16$ volts
max Logic "1" (70%)	< 5.6 volts	< 8.4 volts	< 11.2 volts
min Logic "1" (30%)	> V_{th0}	> V_{th0}	> V_{th0}
max Logic "0" (70%)	< V_{th1}	< V_{th1}	< V_{th1}
min Logic "0" (30%)	> 2.4 volts	> 3.6 volts	> 4.8 volts

The simplest implementation that satisfies the requirements is detection of both logic levels at a fixed threshold -- set at 4.8 volts with no hysteresis employed. At 8 volts V_B , a minimum of 6.4 volts represents a logic "1" at the transmitter -- affording a 1.6 volt margin. Similarly, at 16 volt V_B , a maximum of 3.2 volts represents a logic "0" at the transmitter -- affording a 1.6 volt margin. Enhanced performance may be achieved by the addition of hysteresis and battery tracking of the threshold.



4.3.3 EXAMPLE PHYSICAL LAYER IMPLEMENTATION CIRCUITS

Examples of recommended physical layer implementations for the FORD-9141 network which meet the requirements in this document are shown in Appendix A. Two different sets of physical layer requirements have been defined. The physical layer that is applicable to a particular ECU is dependent on the vehicle implementation and is specified by EESE. Requirements for selecting a particular physical layer are defined in section 4.5. Alternative circuits meeting the requirements defined in this document may be acceptable but require EESE review and approval.

4.4 BUS CHARACTERISTICS

The bus is a single wire that is nominally switched between V_B (battery voltage) and ground. All modules are connected in a wired-OR fashion. In the passive state with a tester connected, the bus "rests" at V_B . With no tester connected, the bus does not rest at any specified voltage, and may vary from zero to V_B , depending on the overall network resistance. The passive voltage state is established by a passive pull-up consisting of multiple parallel resistances within each module on the bus and the tester. The Tester or master unit contains a 510 ohm $\pm 5\%$ pull-up resistance. All other module pull-up resistances are 250K ohm or greater if Physical Layer A is used (Section 4.5.2.1), or they are 500K ohm or greater if Physical Layer B is used (Section 4.5.2.2).

4.5 ELECTRICAL REQUIREMENTS

The communication link shall operate properly over the range 8 volts $< V_B < 16$ volts and within the ground offset conditions specified in Section 4.6. In addition, neither the link nor the module shall be damaged by reverse battery or short of the signal bus wire to 24v @ 60C for up to 1 minute or short to 15v for up to an hour at the module's specified maximum ambient air temperature.

All modules (transmit drivers) shall be capable of withstanding a short of the K-Line to V_B (ranging from -0.7 to 24 Vdc) for an indefinite period of time without failure. Proper operation of the Ford-9141 network is not required in the shorted condition. In order to switch the bus from logic "0" to logic "1" in less than one-tenth of a bit time ($= 9.6$ msec), total bus capacitance shall be limited to less than 9600 picofarads.

The total key-off K-line current on the vehicle shall be less than or equal to 250 microamperes.

4.5.1 TESTER ELECTRICAL REQUIREMENTS

At logic "1", the Tester shall have an equivalent source resistance of 510 ohms $\pm 5\%$.

At logic "0", the Tester shall be able to sink a current of at least 200 mA with V_B at 16 volts. The logic "0" voltage level shall be less than or equal to 1.25 volts $[(16)(10\%) - 0.350 \text{ ground offset}]$.

The Tester shall withstand an indefinite 16 volt K-line to battery short while transmitting.

The Tester and its cable shall represent a capacitance less than 2000 picofarads.

An example of a tester physical layer meeting these requirements is shown in Appendix A.

4.5.2 ECU ELECTRICAL REQUIREMENTS

Two different sets of physical layer requirements have been defined. The physical layer that is applicable to a particular ECU is dependent on the vehicle implementation and is specified by EESE. The physical layer is specified according to the following guidelines. If the number of nodes on the network is less than or equal to five, a maximum of three nodes may use physical layer A (an example is shown in figure 1). All other nodes must use a physical layer meeting the requirements of physical layer B (an example is shown in figure 2). EESE will specify the required physical layer option for a particular ECU. Example physical layers meeting both sets of requirements are shown in Appendix A.



4.5.2.1 ECU PHYSICAL LAYER A

Module pull-up resistances shall be greater than 250K ohms.

Each ECU module input impedance shall be greater than 50K ohms from K-line to ground.

To impress a Logic "0", ECU modules are required to electronically provide a shunt of 110 ohms or less to ground.

Each ECU shall represent a capacitance of no more than 250 picofarads.

4.5.2.2 ECU PHYSICAL LAYER B

Module pull-up resistances shall be greater than 500K ohms. An open circuit from K-line to battery is the preferred pull-up implementation.

Each ECU module input impedance shall be greater than 100K ohms and less than 500K ohms from K-line to ground.

To impress a Logic "0", ECU modules are required to electronically provide a shunt of 110 ohms or less to ground.

Each ECU shall represent a capacitance of no more than 250 picofarads.

4.6 GROUNDING REQUIREMENTS

The FORD-9141 diagnostic communication bus does not require that a separate ground be provided between modules. However, it is important that "high quality grounds" be used at each module. If multiple ground paths exist, the physical interface circuitry shall be referenced to "logic" ground, and not connected to "actuator load return" grounds. The network is designed to operate in the presence of noise signals and voltage offsets of less than 1.0 volt between the offboard Tester and any onboard ECU. This offset voltage can be partitioned to 0.350 volts between the battery ground and the offboard Tester, and 0.650 volts between the battery ground and the onboard ECU.

4.7 ENVIRONMENTAL REQUIREMENTS

Each ECU shall operate on the FORD-9141 communication bus over a temperature range of -40°C to 85°C. Some ECUs may be required to operate over an expanded range of -40°C to 125°C.

Each ECU shall tolerate storage over a temperature range of -55°C to 150°C.

4.8 FAULT TOLERANCE

Any ECU that resets shall continue to meet K-line leakage current and resistance specifications. If any ECU resets, the remaining ECUs shall be capable of network communication.

All ECUs shall continue to meet K-line leakage current and resistance specifications during a loss of power. If any ECU loses its power connection, the remaining ECUs shall remain capable of network communication.

If an ECU has an open power ground, the network may not operate. Network operation may be lost due to a potential AC voltage short to V_b via the blocking diode and a power supply bypass capacitor under open power conditions.

The network will not operate during the following K-line fault conditions:

1. K-line open (dependent on the open location)
2. K-line short to battery
3. K-line short to ground



5.0 MESSAGE STRUCTURE, CONTENT, AND DIALOG REQUIREMENTS

The sections which describe diagnostic message structure, content and dialogue have been deleted from this document and added to DS-YF1T-1A294-AB - "Global Diagnostic Specification". This document now contains only protocol definition and interface requirements.

APPENDIX A SAMPLE PHYSICAL LAYER IMPLEMENTATION CIRCUITS

Figure 1 and Figure 2 and are recommended circuits for meeting the ECU requirements defined in Section 4.5.

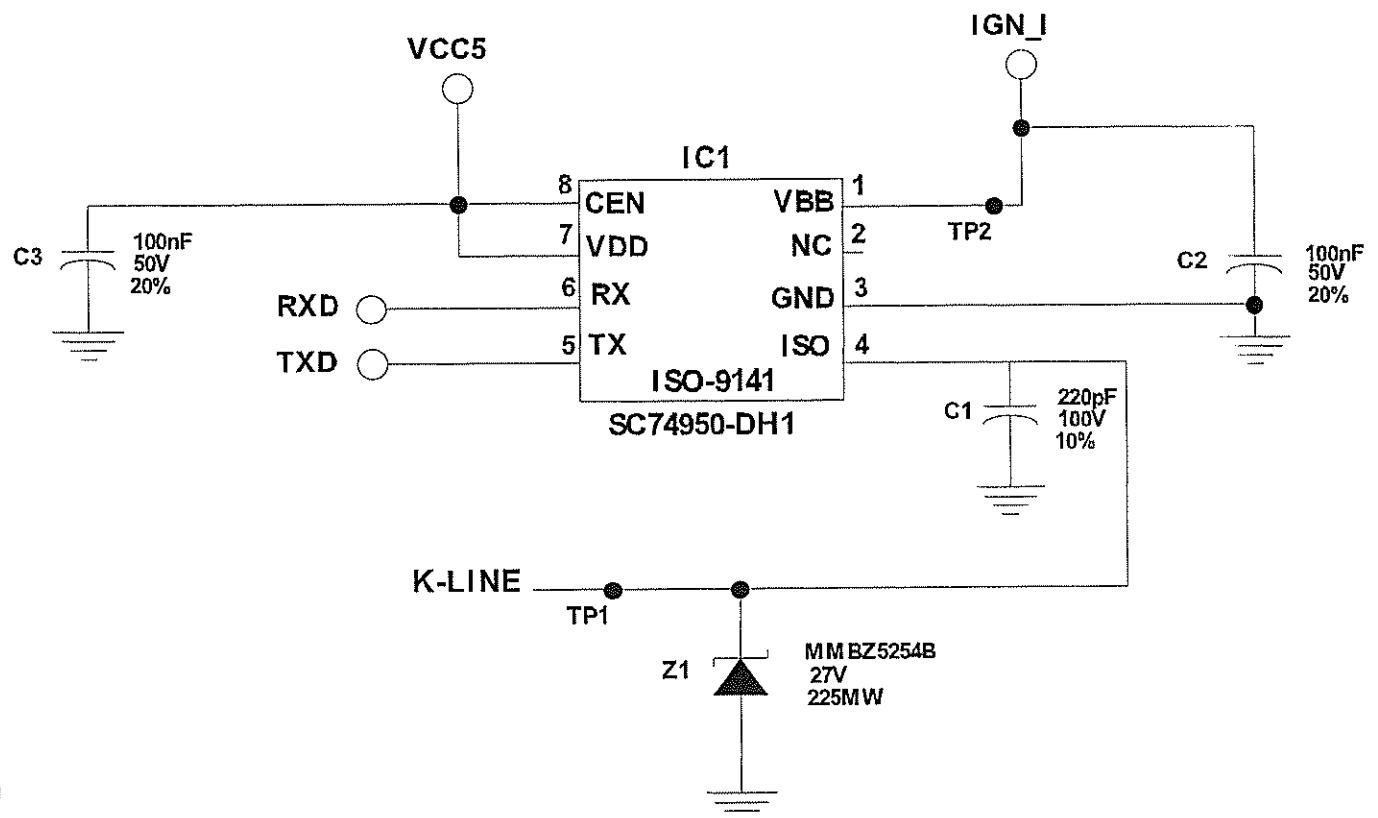
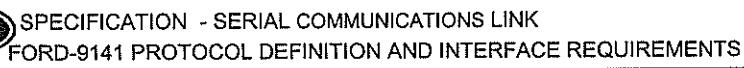


Figure 1 Physical Layer A



REV. FD | FRAME 15 of 20
DECEMBER 16, 1997

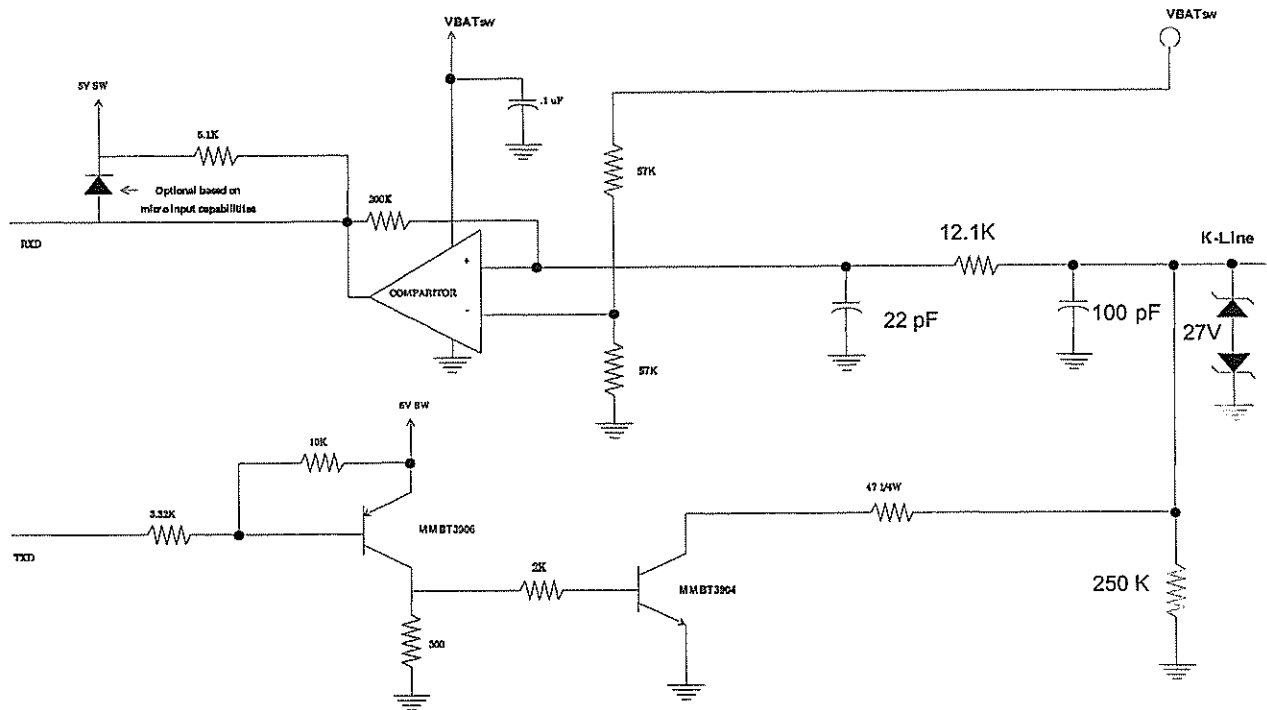


Figure 2 Physical layer B



Figure 3 is a recommended circuit for meeting the Tester requirements defined in Section 4.5.

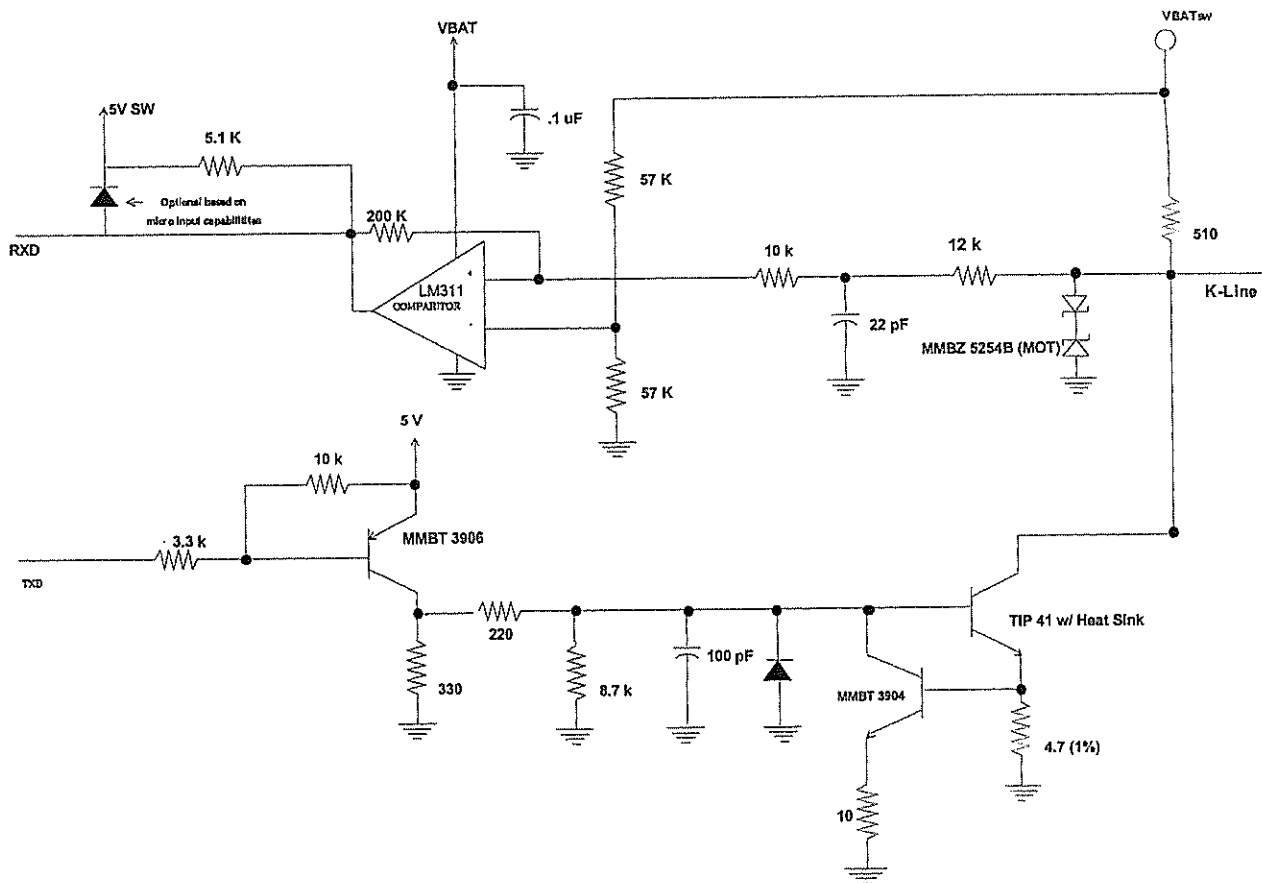


Figure 3 Tester Physical Layer



Design assumptions for the Ford-9141 physical layer;

The following analysis describes the design rationale for choosing the components used in, Physical Layer B., ISO-9141 Drive Circuit. The analysis, provides details for the transmit portion of the Physical Layer B circuit shown in Figure 4.

1. Switching characteristics of Q_1 and Q_2 must be less than 2 milliseconds.
2. The ECU software must protect the output drive transistor Q_1 during a short to battery fault condition. The transistor Q_1 and collector resistor R_C will survive a battery short for one byte time. The software must monitor the clocked out byte and compare it to the received byte. If the bytes don't match, the ECU shall stop transmitting the remaining bytes of the message.
3. R_C temperature compensation must be included in the calculation. The operating temperature is -40°C to 85°C . For special applications, the operating temperature is -40°C to 125°C .
4. The comparator bias supply needs protection from the following electrical conditions:
 - a) Reverse battery
 - b) Load dump
 - c) Non-inverting input can be brought below ground by as much as 1.0 volts.

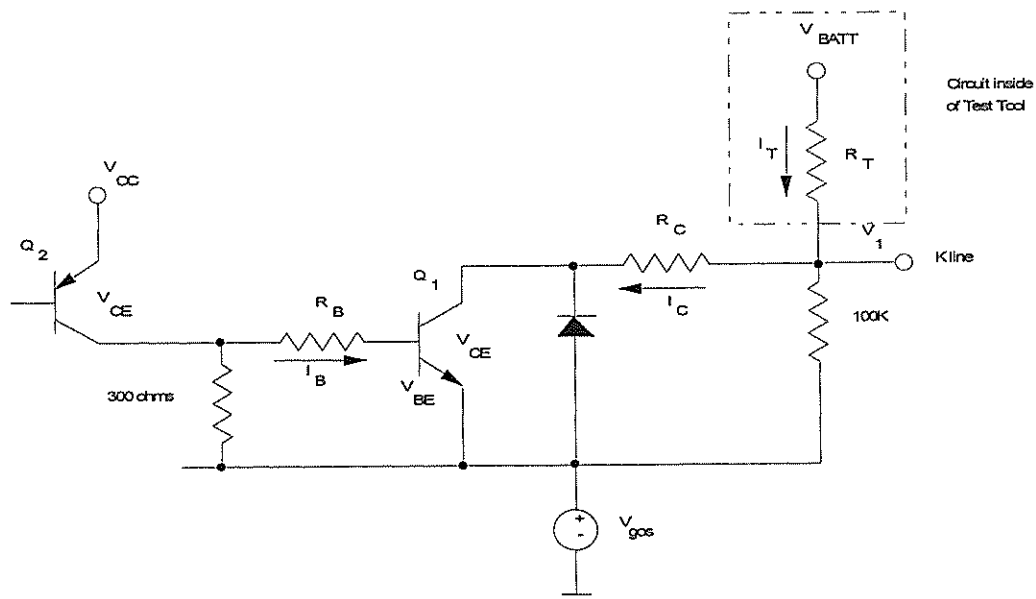


Figure 4 Ford-9141 Drive circuit for Physical Layer B

The maximum operating current (I_T) that will flow through the resistor (R_T) is calculated with the following assumptions;

- a) $V_B = 16.0$ volts
- b) $R_T = 510\text{ohms} - 5\% = 485\text{ohms}$
- c) The 100Kohms resistor is not included in the calculation because $100\text{K} \gg 510$ ohms
- d) $V_{gos} = 0.0$ volts
- e) $V_1 = (16.0 \times 20\%) = 3.2$ volts



$$I_T \leq \frac{V_{BATT} - (V_I - V_{gos})}{R_T} \leq \frac{16.0 - (3.2 - 0.0)}{485} \leq 26.4mA$$

The minimum operating current (I_T) that will flow through the resistor (R_T) is calculated with the following assumptions;

- a) $V_B = 8.0$ volts
- b) $R_T = 510\Omega - 5\% = 485\Omega$
- c) The $100K\Omega$ resistor is not included in the calculation because $100K\Omega \gg 510\Omega$
- d) $V_{gos} = 0.0$ volts
- e) $V_I = (8.0 \times 20\%) = 1.6$ volts

$$I_T \leq \frac{V_{BATT} - (V_I - V_{gos})}{R_T} \leq \frac{8.0 - (1.6 - 0.0)}{485} \leq 13.2mA$$

The maximum value for R_C is calculated with the following assumptions;

- a) $V_B = 8.0$ volts
- b) $V_I = V_B \times 20\% = 1.6$ volts
- c) $V_{CE} = 0.2$ volts
- d) $V_{gos} = 0.65$ volts
- e) $I_C \gg I_T$; because $R_C \ll 100K\Omega$
- f) $I_T = 0.0132$ amps

$$R_C \leq \frac{V_I - V_{gos} - V_{CE}}{I_C} \leq \frac{1.6 - 0.65 - 0.2}{0.0132} \leq 56.8\Omega$$

The recommended value for R_C is 47Ω .

The power dissipated by R_C is calculated with the following assumptions;

- a) $I_{CE} = 0.0264$ amps
- b) $V_B = 16.0$ volts
- c) $V_{CE} = 0.2$ volts

R_C with a tolerance of 5% and a derating factor of 3% (based on an operating temperature of 85°C , surface mounted, carbon resistor), is calculated by $47\Omega \times 1.08 = 50.8W$. The power calculation for R_C is as follows:

$$ResistorPowerRating \geq I_{CE}^2 \times R_C \geq 0.0264^2 \times 56.8 \geq 39.5mW$$

Note: This calculation assumes a D.C. voltage.



The maximum value of R_B is calculated with the following assumptions;

- a) β is 36 @ -40°C
- b) V_{CC} is 5.0 volts $\pm 5\%$ use 4.75 volts
- c) $V_{BE} = 0.7$ volts
- d) $V_{CE} = 0.2$ volts
- e) $I_{C_{MAX}} = 0.0264$ amps
- f) $I_B = I_{C_{MAX}}/\beta$

$$R_B \leq \frac{V_{CC} - V_{CE} - V_{BE}}{I_B} = \frac{4.75 - 0.2 - 0.85}{\frac{I_{C_{MAX}}}{\beta}} = \frac{3.7 \times 36}{0.0264} = 5045\Omega$$

R_B with a tolerance of 5% and 3% for derating, is calculated $5045\Omega * 1.08 = 5449\Omega$; The maximum value of R_B must be less than $5.4\text{K}\Omega$, so the allowable range of R_B is $1.5\text{K}\Omega < R_B < 5.4\text{K}\Omega$. A value of $2.0\text{K}\Omega$ is recommended.



APPENDIX B

CHANGES FROM ES-FL7C-12K529-FC TO ES-FL7C-12K529-FD

GENERAL:	changed document title to: "FORD-9141 PROTOCOL DEFINITION AND INTERFACE REQUIREMENTS"
Page 2:	Added: copyright statement
Section 1:	modified to reflect the revised content of the document which describe only the 9141 protocol and no longer contains specification of message structure, content, or dialog.
Section 1.1:	modified to reflect the revised content of the document which describe only the 9141 protocol and no longer contains specification of message structure, content, or dialog.
Section 1.2:	modified to reflect the revised content of the document which describe only the 9141 protocol and no longer contains specification of message structure, content, or dialog. Added reference to DS-YF1T-1A294-AB - "Global Diagnostic Specification".
Section 2:	deleted reference to "highest" and "lowest" ranking of referenced documents.
Section 4.5:	Modified the first paragraph of section 4.5 to read: "The communication link shall operate properly over the range 8 volts < V_B < 16 volts and within the ground offset conditions specified in Section 4.6. In addition, neither the link nor the module shall be damaged by reverse battery or short of the signal bus wire to 24v @ 60C for up to 1 minute or short to 15v for up to an hour at the module's specified maximum ambient air temperature."
Sections 5, 6, & 7:	These sections which describe diagnostic message structure and dialogue have been deleted from this document and added to DS-YF1T-1A294-AB - "Global Diagnostic Specification". This document now contains only protocol definition and interface requirements.
APPENDICES	Deleted old Appendix A, "DIAGNOSTIC MESSAGE MODE ASSIGNMENT" and old Appendix B, "DIAGNOSTIC MESSAGE RESPONSE CODES". Renamed old Appendix C to Appendix A, and renamed old Appendix D to Appendix B.

LAST PAGE



ENGINEERING SPECIFICATION

PART NAME											SPECIFICATION - SERIAL COMMUNICATIONS LINK							PART NUMBER						
FORD-9141 PROTOCOL DEFINITION AND INTERFACE REQUIREMENTS											ES-F7LC-12K529-FD													
LET																								
FR	1	1A	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16							
LET																								
FR	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33							
DATE	LET	FR	REVISIONS								DR	CK	REFERENCE											
7/26/94			INITIAL RELEASE- NOTICE NUMBER:										PREPARED/APPROVED BY:											
11/15/94																								
5/17/95																								
6/17/96			DRAFT										CHECKED BY:											
9/24/96			Revision "FC" Release																					
12/16/97			"FD" RELEASE																					
													DETAILED BY:											
													CONCURRENCE / APPROVAL SIGNATURES											
													DESIGN ENGINEERING SUPERVISOR											
													DESIGN ENGINEERING											
													MANUFACTURING ENGRG											
													QUALITY CONTROL											
													PURCHASING											
													SUPPLIER QUALITY ASSISTANCE											
													</											

Appendix A MESSAGE STRUCTURES & RESPONSES

A.1 PURPOSE

The first part of this Appendix provides a brief description of the structure of the SCP and ISO-9141-FORD serial communications link Request and Response messages used for Diagnostics.

The second part of this Appendix (section A.6) defines the content of each Diagnostic Request and Response message and specifies the set of valid ECU Responses to Request messages received from a Tester while the ECU is in each of its possible operating states.

Request messages are sent by a Tester to an ECU. They are used to request information from an ECU or to direct the ECU to perform a specific function.

Response messages are returned by an ECU to a Tester. They either contain information requested by a Tester, or notify the Tester of acceptance or rejection of the request.

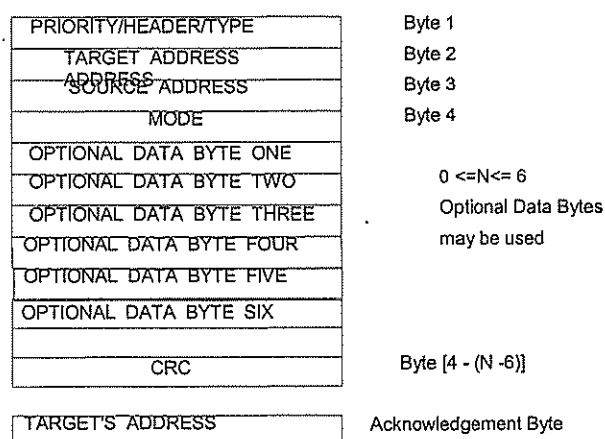


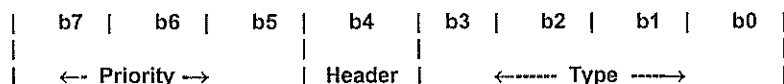
Figure Appendix A -1 SCP Message Structure

A.2 SCP MESSAGE STRUCTURE

All Diagnostic SCP Request and Response messages are structured as shown in Figure A-1. The following subsections briefly define each of the bytes which make up a SCP Message.

A.2.1 SCP Priority / Header / Type Byte

The first byte of a SCP message is the Priority / Header / Type Byte which is defined according to the following bit mapping.



The Priority bits (bit 5 to bit 7) define the importance of a message on the SCP bus for arbitration purposes. The most important message has the highest priority during arbitration for the bus and has the highest priority bit setting. Likewise, the least important message has the lowest priority and the lowest priority bit setting. The correlation between Arbitration Priority and bit settings is as follows:

<u>Bit Setting</u>	<u>Priority</u>
0 0 0	highest
:	:
1 1 0	Diagnostics
1 1 1	lowest

The Header bit (bit 4) distinguishes between 3-Byte Headers (bit 4 = 0) and 1-Byte Headers (bit 4 = 1). All FORD SCP messages implement 3-Byte Headers as follows:

The Type nibble (bit 0 to bit 3) identifies the addressing mode of the message being transmitted. All FORD SCP messages are either Physically or Functionally Addressed. Physically Addressed messages use the unique physical address assigned to the module or Tester as the Target Address of the message being transmitted. Functionally Addressed messages use an address which is unique to a specific function that one or more modules support as the Target Address of the message being transmitted. Since multiple modules may support a specified function, Functional Addressing allows for global communication with several or all modules on an SCP network.

For Diagnostics, an example of a diagnostic broadcast message is **Request No Stored Codes Logging** (Type 1 - broadcast) that has a Type bit setting of [0001]. All Physically Addressed (Type 4 - Node to Node) messages have a Type bit setting of [0100].

A.2.2 SCP Target Address Byte

Byte 2 of a SCP message's Header identifies the Target destination of the message. For Physically Addressed messages, this byte contains the physical address of the Module to which the message is being sent. For Functionally Addressed messages, the byte contains the logical address of the Function to be executed.

A.2.3 SCP Source Address Byte

Header Byte 3 of an SCP message identifies the Physical Address of the Source from which the message originates.

A.2.4 SCP Mode Byte

Byte 4 of an SCP message is the Mode Byte; It uniquely identifies the purpose of the message.

A.2.5 SCP Optional Data Bytes

Following the Mode Byte are up to six Optional Data Bytes. The number of these Bytes and their meaning is specific to a particular message and are defined individually in each of the SCP / ISO-9141-FORD message descriptions.

A.2.6 SCP Cyclic Redundancy Code (CRC) Byte

The last Byte of a SCP message is the Cyclic Redundancy Code (CRC) Byte which is used for message error detection. The algorithm for calculating the CRC can be found in **SAE J1850 - Class B Data Communications Network Interface**.

A.2.7 SCP Acknowledgment Byte

The final Byte shown in Figure A-1 is the SCP Acknowledgment Byte. This Byte is the Physical Address of the Target which is returned by the Target to acknowledge receipt of the message. It does not get transmitted by the Source with the rest of the message. NOTE: To avoid confusion, this Byte will be omitted from the message definitions in this Appendix.

A.3 SCP Diagnostic Description

A.3.1 Qualifying an SCP reception

A SCP diagnostic message response is accepted if the specified target field matches either the physical or functional target(s) of the tester. The tester needs to be addressed physically or have a specific functional address listed in its lookup table for reception of a functional message.

A.3.2 SCP-CARB (California Air Resources Board) Capabilities

The capabilities of the Ford implementation of SCP-CARB are accomplished using only functionally addressed messages. The methodology is being driven by the Society of Automotive Engineers in Recommended Practice SAE J1979 (governing document). The capabilities for SCP-CARB are as listed below:

- Parameter access capability
- Stored codes management capability
- Security access capability

SCP-CARB is intended to conform to government requirements.

A.3.3 Ford SCP Message Timing Requirements

Boundary specifications are necessary to define timely and orderly flow of information on the communication bus. Maximum times are specified to provide a means to automatically terminate reception of aborted or faulty transmissions. Minimums are specified to prevent receiver overrun conditions.

A.3.3.1 Interbyte Gap

Messages are composed of bytes that are sequentially transmitted without interruption. Interbyte gaps are not allowed in the Standard Corporate Protocol.

A.3.3.2 Intermessage Gap

Transactions are composed of messages that are sequentially transmitted. Specification of intermessage gap time sets the maximum and minimum transaction time with a 1 millisecond tolerance. The period between the completion of the last message and the start of the next message is dependent on the source of the messages. Limits are defined:

- 1) ECU RESPONSE FOLLOWING A TESTER REQUEST:
 $0 \leq \text{INTERMESSAGE GAP TIME} \leq 50 \text{ MILLISECONDS}$

This condition occurs when the ECU responds to any Tester query. The Tester must be able to receive messages back-to-back for an indefinite period. In addition, if an expected response is not returned within 75 milliseconds (150% of 50 ms), the Tester times out and may assume that none is forthcoming.

- 2) ECU RESPONSE MESSAGE FOLLOWING ANOTHER ECU MESSAGE IN A SEQUENCE:
 $0 \leq \text{INTERMESSAGE GAP TIME} \leq 50 \text{ MILLISECONDS}$

This condition occurs when an ECU is responding with multiple messages to a single query. The Tester must be able to receive messages back-to-back for an indefinite period. In addition, if the message is not returned within 75 milliseconds (150% of 50 ms), the Tester may time out and shall assume that no more messages will be sent.

- 3) TESTER REQUEST FOLLOWING AN ECU RESPONSE
 $55 \leq \text{INTERMESSAGE GAP TIME} < 5,000 \text{ MILLISECONDS}$

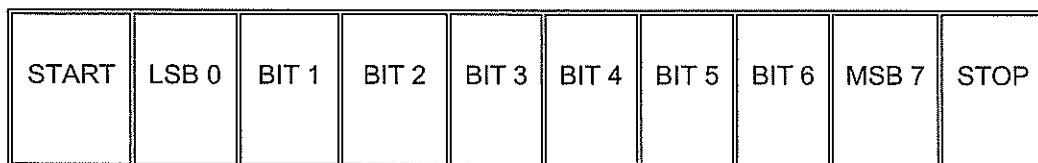
If a response message is not received within 75 milliseconds, the Tester may time out and shall assume that no more messages will be sent. A new Request may be transmitted by the Tester 75 milliseconds (150% of 50 ms) after transmission of the original Request.

The upper limit of 5,000 milliseconds is set to allow an ECU to detect when no Tester is connected to the communication bus. Whenever a Tester is connected, it is required to send a message to each ECU at this minimum rate to continue the diagnostic state dialogue. When an ECU times out, it shall reset and revert to a non-diagnostic state. The 55 millisecond minimum time allows the ECU processing time and prevents the Tester from overrunning the ECU receive buffer. However, an ECU may support PID \$E210, which specifies the shortest intermessage gap the ECU shall tolerate. A value of less than 55 milliseconds may be specified.

NOTE: *Intermessage gap specifications are derived from module response capabilities rather than protocol latency attributes. Increase in protocol speed has little influence on intermessage gap requirements.*

A.4 ISO-9141-FORD MESSAGE STRUCTURE

Information is transferred in groupings of bits, transmitted one byte at a time. The preferred method of implementation employs a Universal Asynchronous Receiver Transmitter (UART) "clockless" technique for transferring bits on a single wire. "Bit banging" is permissible but the bit detection algorithm shall be equivalent to the synchronization and sampling technique commonly applied with UARTs. In accordance with this technique, the receiver shall sample the bus at 16 times the 10.4k bit rate to detect transition of the line from idle and qualify bits with center bit sampling. The format for UART byte transmission is shown below:



The bus normally "rests" at nominal battery voltage (logic "1") when no transmission is occurring. The beginning of a byte transmission is always signaled by the transmission of a Start Bit. This causes the transition of the bus from VB (logic "1") to nominal ground potential (logic "0") for a duration of 1 bit time (96 microseconds). Eight data bits, transmitted least significant bit first, follow the Start Bit. A Stop Bit, always a logic "1", follows the eight data bits. This transmission is commonly referred to as a byte transmission because the start and stop bits are typically "stripped away" and not processed by the CPU.

All ISO-9141-FORD Request and Response messages are structured as shown in Figure A-2.

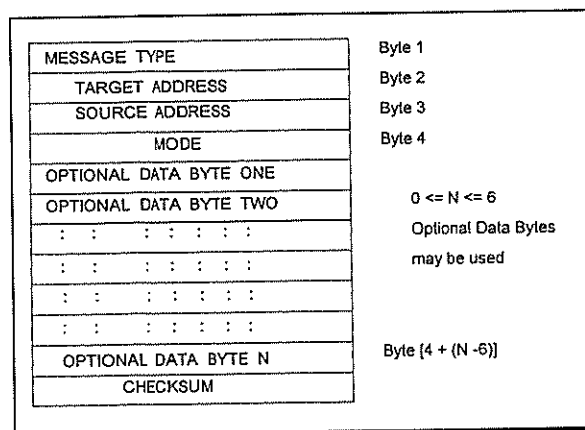


Figure Appendix A -2 ISO-9141-FORD Message Format

The following subsections provide a brief Byte-by-Byte overview of the ISO-9141-FORD message structure.

A.4.1 ISO-9141-FORD Message Type Byte

The first Byte of an ISO-9141-FORD message is the Message Type byte. The Lower Nibble identifies the message "Type". A hex value of '4' indicates that the message is Physically Addressed and a value of '8' indicates that the message is Functionally Addressed. Note: Functionally Addressed messages are not allowed.

The Upper Nibble of the first byte, specifies the "Length" of the Message. The "Length" defines the number of Bytes in the message that follow the Message Type byte including the Checksum. Therefore, the value of the "Length" nibble will be one less than the actual number of bytes that make up the message.

All ISO-9141-FORD Diagnostic messages defined in section A.6 of this Appendix are Physically Addressed and have a Type setting of hex '4'.

A.4.2 ISO-9141-FORD Target Address Byte

Byte 2 of an ISO-9141-FORD message identifies the Physical Address of the Target to which the message is being sent.

A.4.3 ISO-9141-FORD Source Address Byte

Byte 3 of an ISO-9141-FORD message identifies the Physical Address of the Source from which the message originates.

A.4.4 ISO-9141-FORD Mode Byte

Byte 4 of an ISO-9141-FORD message is the Mode byte. It uniquely identifies the purpose of the Request or Response message.

A.4.5 ISO-9141-FORD Optional Data Bytes

Following the Mode byte are Optional Data Bytes. The number of these Bytes (0 - 6) required per message and their meaning is specific to a particular message and are defined individually in each of the ISO-9141-FORD message descriptions.

A.4.6 ISO-9141-FORD Checksum Byte

The last Byte of an ISO-9141-FORD message is the Checksum which is used for Message Error Detection. Its value is the least significant Byte of the sum (with carry) of all of the Bytes in the message excluding the Checksum.

A.4.7 ISO-9141-FORD Message Timing Requirements

Boundary specifications are necessary to define timely and orderly flow of information on the communication bus. Maximum times are specified to provide a means to automatically terminate reception of aborted or faulty transmissions. Minimums are specified to prevent receiver overrun conditions.

A.4.7.1 Interbyte Gap

Messages are composed of sequentially transmitted bytes. For all messages, the period between the completion of the stop bit of one byte and the first edge of the start bit of the next byte is defined. All times specified below include receiver and transmitter tolerances:

- 1) ECU TRANSMISSIONS: $0 \leq \text{INTERBYTE GAP TIME} \leq 20 \text{ MILLISECONDS}$

The ECU may send bytes end-to-end with no gaps. The ECU shall insure that bytes are not separated by more than 20 milliseconds. The ECU shall not transmit late if the byte cannot be sent on time. A maximum length message (11 bytes & 10 maximum gaps) requires approximately 210 milliseconds. The minimum length message (5 bytes and no gaps) requires approximately 5 milliseconds.

- 2) ECU RECEPTION: $3 \leq \text{INTERBYTE GAP TIME} \leq 22 \text{ MILLISECONDS}$

The receiving ECU must receive bytes in the above range. Receiving ECU's shall time-out after 22 milliseconds and assume end-of-message or message abort. A receiving ECU may be capable of receiving messages with interbyte gaps of at least 3 milliseconds. An ECU may support PID \$E210, which specifies an interbyte gap less than the 3 millisecond value specified above. PID \$E210 is defined as two bytes:

- Byte 1: Minimum Interbyte Separation (0 - 255 ms)
- Byte 2: Tester Intermessage Time (0 - 255 ms)

- 3) TESTER TRANSMISSIONS: $5 \leq \text{INTERBYTE GAP TIME} \leq 20 \text{ MILLISECONDS}$

The tester shall not send bytes end-to-end. In most cases, at least 5 milliseconds shall elapse between bytes to prevent overrun of the ECU receive buffer. However, modules may support PID \$E210 which specifies the shortest interbyte gap the ECU may tolerate. The tester shall insure that bytes are not separated by more than 20 milliseconds. Interbyte gaps greater than 22 milliseconds will result in invalid messages. The tester shall not transmit late if the byte cannot be sent on time. A maximum length message (11 bytes & 10 maximum gaps) requires approximately 210 milliseconds. The minimum length message (5 bytes and 4 minimum gaps) requires approximately 25 milliseconds.

- 4) TESTER RECEPTION: $0 \leq \text{INTERBYTE GAP TIME} \leq 22 \text{ MILLISECONDS}$

The tester shall be capable of receiving up to 11 bytes without an interbyte gap. The tester receiver shall time-out after 22 milliseconds and assume end-of-message or message abort.

A.4.7.2 Intermessage Gap

Transactions are composed of messages that are sequentially transmitted. Specification of intermessage gap time sets the maximum and minimum transaction time with a 1 millisecond tolerance. The period between the completion of the last message and the start of the next message is dependent on the source of the messages. Limits are defined:

- 1) ECU RESPONSE FOLLOWING A TESTER REQUEST:
 $0 \leq \text{INTERMESSAGE GAP TIME} \leq 50 \text{ MILLISECONDS}$

This condition occurs when the ECU responds to any Tester query. The Tester must be able to receive messages back-to-back for an indefinite period. In addition, if an expected response is not returned within 75 milliseconds (150% of 50 ms), the Tester times out and may assume that none is forthcoming.

- 2) ECU RESPONSE MESSAGE FOLLOWING ANOTHER ECU MESSAGE IN A SEQUENCE:
 $0 \leq \text{INTERMESSAGE GAP TIME} \leq 50 \text{ MILLISECONDS}$

This condition occurs when an ECU is responding with multiple messages to a single query. The Tester must be able to receive messages back-to-back for an indefinite period. In addition, if the message is not returned within 75 milliseconds (150% of 50 ms), the Tester may time out and shall assume that no more messages will be sent.

- 3) TESTER REQUEST FOLLOWING AN ECU RESPONSE
 $55 \leq \text{INTERMESSAGE GAP TIME} < 5,000 \text{ MILLISECONDS}$

If a response message is not received within 75 milliseconds, the Tester may time out and shall assume that no more messages will be sent. A new Request may be transmitted by the Tester 75 milliseconds (150% of 50 ms) after transmission of the original Request.

The upper limit of 5,000 milliseconds is set to allow an ECU to detect when no Tester is connected to the communication bus. Whenever a Tester is connected, it is required to send a message to each ECU at this minimum rate to continue the diagnostic state dialogue. When an ECU times out, it shall reset and revert to a non-diagnostic state. The 55 millisecond minimum time allows the ECU processing time and prevents the Tester from overrunning the ECU receive buffer. However, an ECU may support PID \$E210, which specifies the shortest intermessage gap the ECU shall tolerate. A value of less than 55 milliseconds may be specified.

NOTE: *Intermessage gap specifications are derived from module response capabilities rather than protocol latency attributes. Increase in protocol speed has little influence on intermessage gap requirements.*

A.5 SCP and ISO-9141-FORD TARGET/SOURCE ADDRESS ASSIGNMENTS

Module and Tester Physical Address assignments reflected in the Target (byte 2) and Source (byte 3) bytes of all Request / Response messages adhere to the following ranges as outlined in **SAE Recommended Practice J2178 - Class B Data Communication Network Messages**.

<u>Target / Source Node Category</u>	<u>Physical Address Range</u>
Powertrain Controllers	00 - 1F
Chassis Controllers	20 - 3F
Body Controllers	40 - C7
Testers	F0 - FD
All Nodes	FE
Null Node	FF

For more in depth information on the Physical Address Ranges assigned to each Node Category and the Range (C8 - EF), reference **SAE J2178**.

NOTE: ECU address assignments are the same across all vehicle lines (e.g., Taurus ABS address = 28h = Explorer ABS address). ECU should respond to any tester address (i.e., no need to filter data to determine if address is within range).

A.6 VALID ECU-TO-TESTER RESPONSES

Each ECU Response that is valid while an ECU is in a particular OPERATING STATE / MODE is identified by name and Mode number under the column heading VALID RESPONSES.

The reason for returning a particular response is listed under the column heading - REASON FOR THE RESPONSE.

To determine the correct ECU Response to a Tester Request message first identify the OPERATING STATE / MODE that the ECU is in when the Request is received and examine the VALID RESPONSES associated with that Operating State. Note: An ECU will ignore any ISO-9141-FORD message that has an invalid length or checksum.

Example 1

During normal operation, an ECU detects failures that cause it to log 4 Continuous DTCs. Later, after being placed in the Diagnostic State by a Tester, the ECU Responds correctly to a PID Request for the Number of Stored Codes (0200). It then receives a **Request Stored Codes** message from the Tester.

What Response should the ECU return ?

During the ECU OPERATING STATE / MODE - Diagnostic State, there is one VALID RESPONSE - **Report Stored Codes [53]**. This Response will be returned to the Tester twice by the ECU because the REASON FOR THE RESPONSE identifies that the ECU will return multiple messages at 3 Continuous DTCs per message until all have been reported. The first **Report Stored Codes [53]** response will return 3 Continuous DTCs in bytes 5 - 10. The second response will return 1 Continuous DTC in bytes 5 & 6 and 00 00 00 00 in bytes 7 - 10. Refer to **Section 9.0, DIAGNOSTIC TROUBLE CODE REQUIREMENTS** for a detailed description of Continuous DTCs and the requirements associated with them.

Example 2

While a vehicle is moving down the road at 60 KPH its ABS module receives a **Request Diagnostic State Entry** message while in its Operational State.

What Response should the ABS Module return ?

During the ECU OPERATING STATE / MODE - Operational State, there are three possible VALID RESPONSES but there is not enough information in the above explanation to formulate a conclusion. In the Subsystem Specific Diagnostic Specification (SSDS Part II) for the ABS Module it states that in order for the ABS Module to enter the Diagnostic State, vehicle speed must be below 10 KPH.

Based on this additional information, the ABS Module should respond with a **General Response [7F] of: [22] Conditions Not Correct** since a prerequisite condition is being violated.

Example 3

Upon being evaluated in the Diagnostic State an ECU receives a **Request Return to Operational State** message to which it responds correctly. 100mS later, a Tester sends the ECU a **Request Parameter by PID** message which contains a PID that is not supported by the ECU.

What Response should the ECU return?

Upon receiving the **Request Return to Operational State** message, while in the Diagnostic State, the ECU has 500 milliseconds during which to reinitialize. During that time period, an ECU is not required to respond to any messages.

What Response should the ECU return if the Tester issues the same Request after waiting 600 milliseconds?

After 500 milliseconds, the ECU is required to respond to Request messages. During the ECU OPERATING STATE / MODE - Operational State (to which the ECU returned), there are three possible VALID RESPONSES. Since the Request contained a PID that is not supported by the ECU (invalid PID specified in bytes 5 & 6), the ECU should respond with a **General Response [7F] of: [12] - Subfunction Not Supported**

A.7 SCP/ISO-9141-FORD DIAGNOSTIC MESSAGES

The following Tables define the content for each Request and Response message used for Diagnostics. It also identifies the complete set of valid ECU Responses to Request messages received from a Tester. Refer to Table A-1 for a complete listing of diagnostic messages.

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REQUEST NO STORED CODES LOGGING [B0] (SCP Only)	A-38
DIAGNOSTIC COMMAND [B1]	A-39

GENERAL RESPONSE [7F]

ECU Response: used to inform the Tester of the acceptance or the rejection (and reason) to a Request message.

Byte-by-Byte Message Description:

BYTE	VALUE	DEFINITION	SCP DESCRIPTION / COMMENTS	ISO-9141-FORD DESCRIPTION / COMMENTS
1	x4	Message Type	Priority/Header/Type (C4)	Length/Type (94)
2	Fx	Target Address	Tester Address	Tester Address
3	xx	Source Address	ECU Address	ECU Address
4	7F	Mode Byte	General Response	General Response
5	xx	Mode Byte of Request Msg	Echo back the Mode byte (byte 4) of the Request Message received	Echo back the Mode byte (byte 4) of the Request Message received
6	xx	Byte 5 of Request Msg	Echo back Byte 5 of Request Msg if Request Msg length > 5 bytes else 00	Echo back Byte 5 of Request Msg if Request Msg length > 5 bytes else 00
7	xx	Byte 6 of Request Msg	Echo back Byte 6 of Request Msg if Request Msg length > 6 bytes else 00	Echo back Byte 6 of Request Msg if Request Msg length > 6 bytes else 00
8	xx	Byte 7 of Request Msg	Echo back Byte 7 of Request Msg if Request Msg length > 7 bytes else 00	Echo back Byte 7 of Request Msg if Request Msg length > 7 bytes else 00
9	xx	Response Code	Identifies the Acceptance / Rejection of the Request (see definitions below)	Identifies the Acceptance / Rejection of the Request (see definitions below)
10	xx	Check Byte	CRC	Checksum

Response Code (byte 9) definitions:

CODE	DEFINITION	MEANING
00	AFFIRMATIVE	The Request received is supported by the module and all data was correct. The ECU will execute the function as requested.
11	MODE NOT SUPPORTED	The ECU does not support the message (Mode).
12	SUB-FUNCTION NOT SUPPORTED, INVALID DATA OR INVALID FORMAT	The ECU supports the message, but arguments contained within the message are not supported, are not correct, or are not in the proper format.
21	BUSY - REPEAT REQUEST	The ECU is currently Busy and is unable to service the request at this time.
22	CONDITIONS NOT CORRECT	The ECU could not service the request because one or more prerequisite conditions required for its support have not been satisfied.
72	TRANSFER ABORTED	Block transfer operation was halted due to some fault, and will not be completed later.

NOTE: If an ECU receives a Mode 7F as a request message, it shall be ignored by the ECU.

REQUEST DIAGNOSTIC STATE ENTRY [10]

Tester Request: that directs the ECU to enter its Diagnostic State.

Byte-by-Byte Message Description:

BYTE	VALUE	DEFINITION	SCP DESCRIPTION / COMMENTS	ISO-9141-FORD DESCRIPTION / COMMENTS
1	x4	Message Type	Priority/Header/Type (Value = C4)	Length/Type (Value = 44)
2	xx	Target Address	ECU Address	ECU Address
3	Fx	Source Address	Tester Address	Tester Address
4	10	Mode Byte	Request Diagnostic State Entry	Request Diagnostic State Entry
5	xx	Check Byte	CRC	Checksum

Valid ECU Responses to: REQUEST DIAGNOSTIC STATE ENTRY:

CURRENT ECU STATE / MODE	VALID RESPONSES	REASON FOR THE RESPONSE	RESPONSE MESSAGE DATA (Byte by Byte)										
			1	2	3	4	5	6	7	8	9	10	11
OPERATIONAL STATE	General Response [7F] of:00 (Affirmative)	- the ECU will be placed into its Diagnostic State	x4	Fx	xx	7F	10	00	00	00	00	xx	
	General Response [7F] of: 21 (Busy)	- the ECU is Busy try again	x4	Fx	xx	7F	10	00	00	00	21	xx	
	General Response [7F] of: 22 (Conditions Not Correct)	- not all Diagnostic State Entry conditions have been satisfied	x4	Fx	xx	7F	10	00	00	00	22	xx	
No Stored Codes Logging Mode	General Response [7F] of:00 (Affirmative)	- the ECU will be placed into its Diagnostic State	x4	Fx	xx	7F	10	00	00	00	00	xx	
	General Response [7F] of: 21 (Busy)	- the ECU is Busy try again	x4	Fx	xx	7F	10	00	00	00	21	xx	
	General Response [7F] of: 22 (Conditions Not Correct)	- not all Diagnostic State Entry conditions have been satisfied	x4	Fx	xx	7F	10	00	00	00	22	xx	
DIAGNOSTIC STATE	General Response [7F] of: 00 (Affirmative)	- the ECU is already in the Diagnostic State	x4	Fx	xx	7F	10	00	00	00	00	xx	
	General Response [7F] of: 22 (Conditions Not Correct)	- not valid while the ECU is executing a Diagnostic Test	x4	Fx	xx	7F	10	00	00	00	22	xx	
	General Response [7F] of: 22 (Conditions Not Correct)	- not valid while the ECU is executing a Diagnostic Command	x4	Fx	xx	7F	10	00	00	00	22	xx	
Memory Transfer Mode	General Response [7F] of: 22 (Conditions Not Correct)	- not valid while the ECU is Transferring Data to/from Memory	x4	Fx	xx	7F	10	00	00	00	22	xx	

REQUEST RETURN TO OPERATIONAL STATE [20]

Tester Request: that directs the ECU to return to its Operational State

Byte-by-Byte Message Description:

BYTE	VALUE	DEFINITION	SCP DESCRIPTION / COMMENTS	ISO-9141-FORD DESCRIPTION / COMMENTS
1	x4	Message Type	Priority/Header Type (C4)	Length/Type (44)
2	xx	Target Address	ECU Address	ECU Address
3	Fx	Source Address	Tester Address	Tester Address
4	20	Mode Byte	Request Return to Operational State	Request Return to Operational State
5	xx	Check Byte	CRC	Checksum

Valid ECU Responses to: REQUEST RETURN TO OPERATIONAL STATE

CURRENT ECU STATE / MODE		VALID RESPONSES	REASON FOR THE RESPONSE	RESPONSE MESSAGE DATA (Byte-by-Byte)										
OPERATIONAL STATE		General Response [7F] of: 00 (Affirmative)	- the ECU is already in the Operational State	x4	Fx	xx	7F	20	00	00	00	00	xx	
No Stored Codes Logging Mode		General Response [7F] of: 00 (Affirmative)	- the ECU will be returned to the Operational State	x4	Fx	xx	7F	20	00	00	00	00	xx	
DIAGNOSTIC STATE		General Response [7F] of: 00 (Affirmative)	- the ECU will be returned to the Operational State *	x4	Fx	xx	7F	20	00	00	00	00	xx	
Diagnostic Test Mode		General Response [7F] of: 00 (Affirmative)	- the ECU will be returned to the Operational State *	x4	Fx	xx	7F	20	00	00	00	00	xx	
Command Mode		General Response [7F] of: 00 (Affirmative)	- the ECU will be returned to the Operational State *	x4	Fx	xx	7F	20	00	00	00	00	xx	
Memory Transfer Mode		General Response [7F] of: 00 (Affirmative)	- the ECU will be returned to the Operational State *	x4	Fx	xx	7F	20	00	00	00	00	xx	

* Note: refer to Section 4.0 - Operational State Diagnostic Requirements for re-initialization timing requirements.

REQUEST STORED CODES [13] (Continuous DTCs)

Tester Request: that directs the ECU to report all Stored Codes (Continuous DTCs) that it currently has logged.

Byte-by-Byte Message Description:

BYTE	VALUE	DEFINITION	SCP DESCRIPTION / COMMENTS	ISO-9141-FORD DESCRIPTION / COMMENTS
1	x4	Message Type	Priority/Header/Type (C4)	Length/Type (44)
2	xx	Target Address	ECU Address	ECU Address
3	Fx	Source Address	Tester Address	Tester Address
4	13	Mode Byte	Request Stored Codes (Continuous DTCs)	Request Stored Codes (Continuous DTCs)
5	xx	Check Byte	CRC	Checksum

Valid ECU Responses to: REQUEST STORED CODES:

	CURRENT ECU STATE / MODE	VALID RESPONSES	REASON FOR THE RESPONSE	RESPONSE MESSAGE DATA (Byte-by-Byte)										
				1	2	3	4	5	6	7	8	9	10	11
OPERATIONAL STATE		Report Stored Codes [53]	- returns all Continuous DTCs logged by the ECU (3 per message)	x4	Fx	xx	53	xx	xx	xx	xx	xx	xx	xx
		General Response [7F] of: 21 (Busy)	- the ECU is Busy with normal operation - try again	x4	Fx	xx	7F	13	00	00	00	21	xx	xx
		Report Stored Codes [53]	- returns all Continuous DTCs logged by the ECU (3 per message)	x4	Fx	xx	53	xx	xx	xx	xx	xx	xx	xx
No Stored Codes Logging Mode		General Response [7F] of: 21 (Busy)	- the ECU is Busy with normal operation - try again	x4	Fx	xx	7F	13	00	00	00	21	xx	xx
		Report Stored Codes [53]	- returns all Continuous DTCs logged by the ECU (3 per message)	x4	Fx	xx	53	xx	xx	xx	xx	xx	xx	xx
		Report Stored Codes [53]	- the ECU is Busy - try again	x4	Fx	xx	7F	13	00	00	00	21	xx	xx
DIAGNOSTIC STATE		General Response [7F] of: 21 (Busy)	- the ECU is Busy - try again	x4	Fx	xx	7F	13	00	00	00	21	xx	xx
		General Response [7F] of: 22 (Conditions Not Correct)	- not valid while the ECU is executing a Diagnostic Test	x4	Fx	xx	7F	13	00	00	00	22	xx	xx
		General Response [7F] of: 22 (Conditions Not Correct)	- not valid while the ECU is executing a Diagnostic Command	x4	Fx	xx	7F	13	00	00	00	22	xx	xx
Command Mode		General Response [7F] of: 22 (Conditions Not Correct)	- not valid while the ECU is Transferring Data to/from Memory	x4	Fx	xx	7F	13	00	00	00	22	xx	xx

REPORT STORED CODES [53]

ECU Response: that is used to report the Stored Codes (Continuous DTCs) currently logged by the ECU (The ECU will return consecutive messages with up to 3 Continuous DTCs per message until all of the Continuous DTCs have been reported).

Byte-by-Byte Message Description:

BYTE	VALUE	DEFINITION	SCP DESCRIPTION / COMMENTS	ISO-9141-FORD DESCRIPTION / COMMENTS
1	x4	Message Type	Priority/HeaderType (C4)	Length/Type (A4)
2	Fx	Target Address	Tester Address	Tester Address
3	xx	Source Address	ECU Address	ECU Address
4	53	Mode Byte	Report Stored Codes	Report Stored Codes
5	xx	MSB of Stored Code	00 if no Continuous DTCs were logged by the ECU	00 if no Continuous DTCs were logged by the ECU
6	xx	LSB of Stored Code	00 if no Continuous DTCs were logged by the ECU	00 if no Continuous DTCs were logged by the ECU
7	xx	MSB of Stored Code	00 if less than 2 Continuous DTCs transferred for this response	00 if less than 2 Continuous DTCs transferred for this response
8	xx	LSB of Stored Code	00 if less than 2 Continuous DTCs transferred for this response	00 if less than 2 Continuous DTCs transferred for this response
9	xx	MSB of Stored Code	00 if less than 3 Continuous DTCs transferred for this response	00 if less than 3 Continuous DTCs transferred for this response
10	xx	LSB of Stored Code	00 if less than 3 Continuous DTCs transferred for this response	00 if less than 3 Continuous DTCs transferred for this response
11	xx	Check Byte	CRC	Checksum

REQUEST CLEAR STORED CODES [14]

Tester Request: that directs the ECU to clear all of the Continuous DTCs that it currently has logged.

Byte-by-Byte Message Description:

BYTE	VALUE	DEFINITION	SCP DESCRIPTION / COMMENTS	ISO-9141-FORD DESCRIPTION / COMMENTS
1	x4	Message Type	Priority/Header/Type (C4)	Length/Type (44)
2	xx	Target Address	ECU Address	ECU Address
3	Fx	Source Address	Tester Address	Tester Address
4	14	Mode Byte	Request Clear Stored Codes (clear all Continuous DTCs)	Request Clear Stored Codes (clear all Continuous DTCs)
5	xx	Check Byte	CRC	Checksum

Valid ECU Responses to: REQUEST CLEAR STORED CODES:

CURRENT ECU STATE / MODE	VALID RESPONSES	REASON FOR THE RESPONSE	RESPONSE MESSAGE DATA (Byte-by-Byte)										
			1	2	3	4	5	6	7	8	9	10	11
OPERATIONAL STATE	General Response [7F] of: 00 (Affirmative)	- all Continuous DTC(s) shall be cleared	x4	Fx	xx	7F	14	xx	xx	00	00	xx	
	General Response [7F] of: 21 (Busy)	- the ECU is Busy with normal operation - try again	x4	Fx	xx	7F	14	xx	xx	00	21	xx	
	No Stored Codes Logging Mode	- all Continuous DTC(s) shall be cleared	x4	Fx	xx	7F	14	xx	xx	00	00	xx	
DIAGNOSTIC STATE	General Response [7F] of: 21 (Busy)	- the ECU is Busy with normal operation - try again	x4	Fx	xx	7F	14	xx	xx	00	21	xx	
	General Response [7F] of: 00 (Affirmative)	- all Continuous DTC(s) shall be cleared	x4	Fx	xx	7F	14	xx	xx	00	00	xx	
	General Response [7F] of: 21 (Busy)	- the ECU is Busy - try again	x4	Fx	xx	7F	14	xx	xx	00	21		
Diagnostic Test Mode	General Response [7F] of: 22 (Conditions Not Correct)	- not valid while the ECU is executing a Diagnostic Test	x4	Fx	xx	7F	14	xx	xx	00	22	xx	
Command Mode	General Response [7F] of: 22 (Conditions Not Correct)	- not valid while the ECU is executing a Diagnostic Command	x4	Fx	xx	7F	14	xx	xx	00	22	xx	
Memory Transfer Mode	General Response [7F] of: 22 (Conditions Not Correct)	- not valid while the ECU is Transferring Data to/from Memory	x4	Fx	xx	7F	14	xx	xx	00	22	xx	

Tester Request: that directs the ECU to return information specified by the logical Parameter Identifier (PID) address.

Byte-by-Byte Message Description:

BYTE	VALUE	DEFINITION	SCP DESCRIPTION / COMMENTS	ISO-9141-FORD DESCRIPTION / COMMENTS
1	x4	Message Type	Priority/Header/Type (C4)	Length/Type (64)
2	xx	Target Address	ECU Address	ECU Address
3	FX	Source Address	Tester Address	Tester Address
4	22	Mode Byte	Request Parameter by PID	Request Parameter by PID
5	xx	PID (MSB)	MSB of the logical Parameter Identifier address	MSB of the logical Parameter Identifier address
6	xx	PID (LSB)	LSB of the logical Parameter Identifier address	LSB of the logical Parameter Identifier address
7	xx	Check Byte	CRC	Checksum

[illegible]

CURRENT ECU STATE /MODE		VALID RESPONSES	REASON FOR THE RESPONSE	RESPONSE MESSAGE DATA (Byte-by-Byte)										
				1	2	3	4	5	6	7	8	9	10	11
OPERATIONAL STATE	Report Parameter by PID [62]		- returns the information identified by the logical PID Address	x4	Fx	xx	62	xx	xx	xx	xx	xx	xx	xx
	General Response [7F] of: 12 (Subfunction Not Supported)		- invalid PID specified in bytes 5 & 6	x4	Fx	xx	7F	22	xx	xx	00	12	xx	
	General Response [7F] of: 21 (Busy)		- the ECU is Busy with normal operation - try again	x4	Fx	xx	7F	22	xx	xx	00	21	xx	
No Stored Codes Logging Mode	Report Parameter by PID [62]		- returns the information identified by the logical PID Address	x4	Fx	xx	62	xx	xx	xx	xx	xx	xx	
	General Response [7F] of: 12 (Subfunction Not Supported)		- invalid PID specified in bytes 5 & 6	x4	Fx	xx	7F	22	xx	xx	00	12	xx	
	General Response [7F] of: 21 (Busy)		- the ECU is Busy with normal operation - try again	x4	Fx	xx	7F	22	xx	xx	00	21	xx	
DIAGNOSTIC STATE	Report Parameter by PID [62]		- returns the information identified by the logical PID Address	x4	Fx	xx	62	xx	xx	xx	xx	xx	xx	
	General Response [7F] of: 12 (Subfunction Not Supported)		- invalid PID specified in bytes 5 & 6	x4	Fx	xx	7F	22	xx	xx	00	12	xx	
	General Response [7F] of: 21 (Busy)		- the ECU is Busy - try again	x4	Fx	xx	7F	22	xx	xx	00	21	xx	
Diagnostic Test Mode	Report Parameter by PID [62]		- returns the information identified by the logical PID Address	x4	Fx	xx	62	xx	xx	xx	xx	xx	xx	
	General Response [7F] of: 12 (Subfunction Not Supported)		- invalid PID specified in bytes 5 & 6	x4	Fx	xx	7F	22	xx	xx	00	12	xx	
	General Response [7F] of: 21 (Busy)		- the ECU is Busy - try again	x4	Fx	xx	7F	22	xx	xx	00	21	xx	
Command Mode	Report Parameter by PID [62]		- returns the information identified by the logical PID Address	x4	Fx	xx	62	xx	xx	xx	xx	xx	xx	
	General Response [7F] of: 12 (Subfunction Not Supported)		- invalid PID specified in bytes 5 & 6	x4	Fx	xx	7F	22	xx	xx	00	12	xx	
	General Response [7F] of: 21 (Busy)		- the ECU is Busy - try again	x4	Fx	xx	7F	22	xx	xx	00	21	xx	
Memory Transfer Mode	Report Parameter by PID [62]		- returns the information identified by the logical PID Address	x4	Fx	xx	62	xx	xx	xx	xx	xx	xx	
	General Response [7F] of: 12 (Subfunction Not Supported)		- invalid PID specified in bytes 5 & 6	x4	Fx	xx	7F	22	xx	xx	00	12	xx	
	General Response [7F] of: 21 (Busy)		- the ECU is Busy - try again	x4	Fx	xx	7F	22	xx	xx	00	21	xx	
	Report Parameter by PID [62]		- returns the information identified by the logical PID Address	x4	Fx	xx	62	xx	xx	xx	xx	xx	xx	
	General Response [7F] of: 12 (Subfunction Not Supported)		- invalid PID specified in bytes 5 & 6	x4	Fx	xx	7F	22	xx	xx	00	12	xx	
	General Response [7F] of: 21 (Busy)		- the ECU is Busy - try again	x4	Fx	xx	7F	22	xx	xx	00	21	xx	

REPORT PARAMETER BY PID [62]

ECU Response: that is used to report information identified by a logical PID Address to the Tester.

Byte-by-Byte Message Description:

BYTE	VALUE	DEFINITION	SCP DESCRIPTION / COMMENTS	ISO-9141-FORD DESCRIPTION / COMMENTS
1	x4	Message Type	Priority/Header/Type (C4)	Length/Type (x4)
2	Fx	Target Address	Tester Address	Tester Address
3	xx	Source Address	ECU Address	ECU Address
4	62	Mode Byte	Report Parameter by PID	Report Parameter by PID
5	xx	MSB of PID	MSB of the logical PID address	MSB of the logical PID address
6	xx	LSB of PID	LSB of the logical PID address	LSB of the logical PID address
7	xx	PID Data	1st byte of data as identified by the logical PID Address	1st byte of data as identified by the logical PID Address
8	xx	PID Data Optional Byte	2nd byte of data or omitted from the message	2nd byte of data or omitted from the message
9	xx	PID Data Optional Byte	3rd byte of data or omitted from the message	3rd byte of data or omitted from the message
10	xx	PID Data Optional Byte	4th byte of data or omitted from the message	4th byte of data or omitted from the message
11	xx	Check Byte	CRC	Checksum

REQUEST PARAMETER BY DMR [23]

Tester Request: that directs the ECU to return data located at the specified DMR ECU physical address.

Byte-by-Byte Message Description:

BYTE	VALUE	DEFINITION	SCP DESCRIPTION / COMMENTS	ISO-9141-FORD DESCRIPTION / COMMENTS
1	x4	Message Type	Priority/Header Type (C4)	Length/Type (84)
2	xx	Target Address	ECU Address	ECU Address
3	xx	Source Address	Tester Address	Tester Address
4	23	Mode Byte	Request Parameter by DMR	Request Parameter by DMR
5	xx	Memory Address MSB (4)	MSB of the ECU Physical Address from which the data will be retrieved	MSB of the ECU Physical Address from which the data will be retrieved
6	xx	Memory Address Byte 3	3rd Byte of the ECU Physical Address from which the data will be retrieved	3rd Byte of the ECU Physical Address from which the data will be retrieved
7	xx	Memory Address Byte 2	2nd Byte of the ECU Physical Address from which the data will be retrieved	2nd Byte of the ECU Physical Address from which the data will be retrieved
8	xx	Memory Address LSB (1)	LSB of the ECU Physical Address from which the data will be retrieved	LSB of the ECU Physical Address from which the data will be retrieved
9	xx	Check Byte	CRC	Checksum

Valid ECU Responses to: REQUEST PARAMETER BY DMR

CURRENT MODE	ECU STATE	INVALID RESPONSES	REASON FOR THE RESPONSE	RESPONSE MESSAGE DATA (Byte-by-Byte)										
OPERATIONAL STATE		Report Parameter by DMR [63]	- returns data beginning at the specified ECU Physical Address	x4	Fx	xx	63	xx	xx	xx	xx	xx	xx	xx
		General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this Mode (message)	x4	Fx	xx	7F	23	xx	xx	xx	11	xx	
		General Response [7F] of: 12 (Subfunction Not Supported)	- Invalid Address in bytes 5, 6 & 7	x4	Fx	xx	7F	23	xx	xx	xx	12	xx	
		General Response [7F] of: 21 (Busy)	- the ECU is Busy with normal operation - try again	x4	Fx	xx	7F	23	xx	xx	xx	21	xx	
		Report Parameter by DMR [63]	- returns data beginning at the specified ECU Physical Address	x4	Fx	xx	63	xx	xx	xx	xx	xx	xx	
No Stored Codes Logging Mode		General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this Mode (message)	x4	Fx	xx	7F	23	xx	xx	xx	11	xx	
		General Response [7F] of: 12 (Subfunction Not Supported)	- Invalid Address in bytes 5, 6 & 7	x4	Fx	xx	7F	23	xx	xx	xx	12	xx	
		General Response [7F] of: 21 (Busy)	- the ECU is Busy with normal operation - try again	x4	Fx	xx	7F	23	xx	xx	xx	21	xx	
		Report Parameter by DMR [63]	- returns data beginning at the specified ECU Physical Address	x4	Fx	xx	63	xx	xx	xx	xx	xx	xx	
		General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this Mode (message)	x4	Fx	xx	7F	23	xx	xx	xx	11	xx	
DIAGNOSTIC STATE		Report Parameter by DMR [63]	- returns data beginning at the specified ECU Physical Address	x4	Fx	xx	63	xx	xx	xx	xx	xx	xx	
		General Response [7F] of: 11 (Mode Not Supported)	- Invalid Address in bytes 5, 6 & 7	x4	Fx	xx	7F	23	xx	xx	xx	11	xx	
		General Response [7F] of: 12 (Subfunction Not Supported)	- the ECU is Busy - try again	x4	Fx	xx	7F	23	xx	xx	xx	12	xx	
		General Response [7F] of: 21 (Busy)	- returns data beginning at the specified ECU Physical Address	x4	Fx	xx	63	xx	xx	xx	xx	xx	xx	
		Report Parameter by DMR [63]	- the ECU does not support this Mode (message)	x4	Fx	xx	7F	23	xx	xx	xx	21	xx	
Diagnostic Test Mode		General Response [7F] of: 11 (Mode Not Supported)	- Invalid Address in bytes 5, 6 & 7	x4	Fx	xx	7F	23	xx	xx	xx	11	xx	
		General Response [7F] of: 12 (Subfunction Not Supported)	- the ECU is Busy - try again	x4	Fx	xx	7F	23	xx	xx	xx	12	xx	
		General Response [7F] of: 21 (Busy)	- returns data beginning at the specified ECU Physical Address	x4	Fx	xx	63	xx	xx	xx	xx	xx	xx	
		Report Parameter by DMR [63]	- the ECU does not support this Mode (message)	x4	Fx	xx	7F	23	xx	xx	xx	21	xx	
		General Response [7F] of: 11 (Mode Not Supported)	- Invalid Address in bytes 5, 6 & 7	x4	Fx	xx	7F	23	xx	xx	xx	11	xx	
Command Mode		General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this Mode (message)	x4	Fx	xx	7F	23	xx	xx	xx	11	xx	
		General Response [7F] of: 12 (Subfunction Not Supported)	- Invalid Address in bytes 5, 6 & 7	x4	Fx	xx	7F	23	xx	xx	xx	12	xx	
		General Response [7F] of: 21 (Busy)	- the ECU is Busy - try again	x4	Fx	xx	7F	23	xx	xx	xx	21	xx	
		Report Parameter by DMR [63]	- returns data beginning at the specified ECU Physical Address	x4	Fx	xx	63	xx	xx	xx	xx	xx	xx	
		General Response [7F] of: 11 (Mode Not Supported)	- Invalid Address in bytes 5, 6 & 7	x4	Fx	xx	7F	23	xx	xx	xx	11	xx	
Memory Transfer Mode		General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this Mode (message)	x4	Fx	xx	7F	23	xx	xx	xx	11	xx	
		General Response [7F] of: 12 (Subfunction Not Supported)	- Invalid Address in bytes 5, 6 & 7	x4	Fx	xx	7F	23	xx	xx	xx	12	xx	
		General Response [7F] of: 21 (Busy)	- the ECU is Busy - try again	x4	Fx	xx	7F	23	xx	xx	xx	21	xx	
		Report Parameter by DMR [63]	- returns data beginning at the specified ECU Physical Address	x4	Fx	xx	63	xx	xx	xx	xx	xx	xx	
		General Response [7F] of: 11 (Mode Not Supported)	- Invalid Address in bytes 5, 6 & 7	x4	Fx	xx	7F	23	xx	xx	xx	11	xx	

REPORT PARAMETER BY DMR [63]

ECU Response: that is used to report 4 bytes of data to the Tester beginning at the specified DMR ECU Physical Address.

Byte-by-Byte Message Description:

BYTE	VALUE	DEFINITION	SCP DESCRIPTION / COMMENTS	ISO-9141-FORD DESCRIPTION / COMMENTS
1	x4	Message Type	Priority/Header/Type (C4)	Length/Type (A4)
2	Fx	Target Address	Tester Address	Tester Address
3	xx	Source Address	ECU Address	ECU Address
4	63	Mode Byte	Report Parameter by DMR	Report Parameter by DMR
5	xx	Memory Address byte 2	2nd Byte of the DMR ECU Physical Address echoed back	2nd Byte of the DMR ECU Physical Address echoed back
6	xx	Memory Address LSB	LSB of the DMR ECU Physical Address echoed back	LSB of the DMR ECU Physical Address echoed back
7	xx	Data from Address	Data from the specified DMR ECU Physical Address	Data from the specified DMR ECU Physical Address
8	xx	Data from Address + 1	Data from the specified DMR ECU Physical Address + 1	Data from the specified DMR ECU Physical Address + 1
9	xx	Data from Address + 2	Data from the specified DMR ECU Physical Address + 2	Data from the specified DMR ECU Physical Address + 2
10	xx	Data from Address + 3	Data from the specified DMR ECU Physical Address + 3	Data from the specified DMR ECU Physical Address + 3
11	xx	Check Byte	CRC	Checksum

REQUEST PARAMETER SCALING/MASKING [24]

Tester Request: that directs the ECU to return scaling or masking information relating to a supported PID.

Byte-by-Byte Message Description:

BYTE	VALUE	DEFINITION	SCP DESCRIPTION / COMMENTS	ISO-9141-FORD DESCRIPTION / COMMENTS
1	x4	Message Type	Priority/Header/Type (C4)	Length/Type (84)
2	xx	Target Address	ECU Address	ECU Address
3	Fx	Source Address	Tester Address	Tester Address
4	24	Mode Byte	Request Parameter scaling/masking information	Request Parameter scaling/masking information
5	xx	Table offset position	Set to \$FF if PID reference used in bytes 6 and 7 instead	Set to \$FF if PID reference used in bytes 6 and 7 instead
6	xx	High byte PID reference	Set to \$FF if table offset in byte 5 used	Set to \$FF if table offset in byte 5 used
7	xx	Low byte PID reference	Set to \$FF if table offset in byte 5 used	Set to \$FF if table offset in byte 5 used
9	xx	Check Byte	CRC	Checksum

Valid ECU Responses to: REQUEST PARAMETER SCALING/MASKING

CURRENT ECU STATE	MODE	INVALID RESPONSES	REASON FOR THE RESPONSE	RESPONSE MESSAGE DATA (Byte-by-Byte)										
				1	2	3	4	5	6	7	8	9	10	11
OPERATIONAL STATE		Report Parameter scaling/masking [64]	- Returns the parameter scaling/masking information	x4	Fx	xx	64	xx	xx	xx	xx	xx	xx	xx
		General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this Mode (message)	x4	Fx	xx	7F	23	xx	xx	xx	11	xx	
		General Response [7F] of: 12 (Subfunction Not Supported)	- invalid Address in bytes 6 & 7	x4	Fx	xx	7F	23	xx	xx	xx	12	xx	
		General Response [7F] of: 21 (Busy)	- the ECU is Busy with normal operation - try again	x4	Fx	xx	7F	23	xx	xx	xx	21	xx	
No Stored Codes Logging Mode		Report Parameter scaling/masking [64]	- Returns the parameter scaling/masking information	x4	Fx	xx	64	xx	xx	xx	xx	xx	xx	xx
		General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this Mode (message)	x4	Fx	xx	7F	23	xx	xx	xx	11	xx	
		General Response [7F] of: 12 (Subfunction Not Supported)	- invalid Address in bytes 6 & 7	x4	Fx	xx	7F	23	xx	xx	xx	12	xx	
		General Response [7F] of: 21 (Busy)	- the ECU is Busy with normal operation - try again	x4	Fx	xx	7F	23	xx	xx	xx	21	xx	
DIAGNOSTIC STATE		Report Parameter scaling/masking [64]	- Returns the parameter scaling/masking information	x4	Fx	xx	64	xx	xx	xx	xx	xx	xx	xx
		General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this Mode (message)	x4	Fx	xx	7F	23	xx	xx	xx	11	xx	
		General Response [7F] of: 12 (Subfunction Not Supported)	- invalid Address in bytes 6 & 7	x4	Fx	xx	7F	23	xx	xx	xx	12	xx	
		General Response [7F] of: 21 (Busy)	- the ECU is Busy - try again	x4	Fx	xx	7F	23	xx	xx	xx	21	xx	
Diagnostic Test Mode		Report Parameter scaling/masking [64]	- Returns the parameter scaling/masking information	x4	Fx	xx	64	xx	xx	xx	xx	xx	xx	xx
		General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this Mode (message)	x4	Fx	xx	7F	23	xx	xx	xx	11	xx	
		General Response [7F] of: 12 (Subfunction Not Supported)	- invalid Address in bytes 6 & 7	x4	Fx	xx	7F	23	xx	xx	xx	12	xx	
		General Response [7F] of: 21 (Busy)	- the ECU is Busy - try again	x4	Fx	xx	7F	23	xx	xx	xx	21	xx	
Command Mode		Report Parameter scaling/masking [64]	- Returns the parameter scaling/masking information	x4	Fx	xx	64	xx	xx	xx	xx	xx	xx	xx
		General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this Mode (message)	x4	Fx	xx	7F	23	xx	xx	xx	11	xx	
		General Response [7F] of: 12 (Subfunction Not Supported)	- invalid Address in bytes 6 & 7	x4	Fx	xx	7F	23	xx	xx	xx	12	xx	
		General Response [7F] of: 21 (Busy)	- the ECU is Busy - try again	x4	Fx	xx	7F	23	xx	xx	xx	21	xx	
Memory Transfer Mode		Report Parameter scaling/masking [64]	- Returns the parameter scaling/masking information	x4	Fx	xx	64	xx	xx	xx	xx	xx	xx	xx
		General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this Mode (message)	x4	Fx	xx	7F	23	xx	xx	xx	11	xx	
		General Response [7F] of: 12 (Subfunction Not Supported)	- invalid Address in bytes 6 & 7	x4	Fx	xx	7F	23	xx	xx	xx	12	xx	
		General Response [7F] of: 21 (Busy)	- the ECU is Busy - try again	x4	Fx	xx	7F	23	xx	xx	xx	21	xx	

REPORT PARAMETER SCALING/MASKING [64]

ECU Response: Used to report parameter scaling/masking information. See chapter 8 section "PID SCALING AND MASKING CAPABILITY" for more information

Byte-by-Byte Message Description:

BYTE	VALUE	DEFINITION	SCP DESCRIPTION / COMMENTS	ISO-9141-FORD DESCRIPTION / COMMENTS
1	x4	Message Type	Priority/HeaderType (C4)	LengthType (A4)
2	Fx	Target Address	Tester Address	Tester Address
3	xx	Source Address	ECU Address	ECU Address
4	64	Mode Byte	Report parameter scaling/masking	Report Parameter by DMR
5	xx	Table offset position	Set to \$FF if PID reference used in bytes 6 and 7 instead	Set to \$FF if PID reference used in bytes 6 and 7 instead
6	xx	High byte PID reference	Set to \$FF if table offset in byte 5 used	Set to \$FF if table offset in byte 5 used
7	xx	Low byte PID reference	Set to \$FF if table offset in byte 5 used	Set to \$FF if table offset in byte 5 used
8	xx	Parameter information byte	Specifies the parameter encoding type and number of bytes in parameter.	Specifies the parameter encoding type and number of bytes in parameter.
9	xx	PID scaling/mask	Signed and unsigned scaling byte or bit packed mask byte. Set to \$00 if scaling not supported for data type	Signed and unsigned scaling byte or bit packed mask byte. Set to \$00 if scaling not supported for data type
10	xx	Mask	Is used if multiple bit mapped data bytes are reported, not used otherwise	Is used if multiple bit mapped data bytes are reported, not used otherwise
11	xx	Check Byte	CRC	Checksum

REQUEST SECURITY ACCESS [27]

Tester Request: that directs the ECU to return a Security Seed Code or to Grant or Deny Access based on a Security Key computed by the Tester from a Seed Code supplied previously by the ECU.

Byte-by-Byte Message Description:

BYTE	VALUE	DEFINITION	SCP DESCRIPTION / COMMENTS	ISO-9141/FORD DESCRIPTION / COMMENTS
1	x4	Message Type	Priority/Header Type (C4)	Length/Type (74)
2	xx	Target Address	ECU Address	ECU Address
3	Fx	Source Address	Tester Address	Tester Address
4	27	Mode Byte	Request Security Access	Request Security Access
5	xx	Security Sub-Mode	01 - Request for Access Seed, 02 - Submit Security Key	01 - Request for Access Seed, 02 - Submit Security Key
6	xx	Security Data Byte 1	If Sec. Sub-Mode - 01: byte 6 - \$00 byte 7 - \$00	If Sec. Sub-Mode - 01: byte 6 - \$00 byte 7 - \$00
7	xx	Security Data Byte 2	If Sec. Sub-Mode - 02: byte 6 - Security Key MSB byte 7 - Security Key LSB	If Sec. Sub-Mode - 02: byte 6 - Security Key MSB byte 7 - Security Key LSB
8	xx	Check Byte	CRC	Checksum

Valid ECU Responses to: REQUEST SECURITY ACCESS:

CURRENT ECU STATE / MODE	VALID RESPONSES	REASON FOR THE RESPONSE	RESPONSE MESSAGE DATA (Byte-by-Byte)										
			1	2	3	4	5	6	7	8	9	10	11
OPERATIONAL STATE	General Response [7F] of: 22 (Conditions Not Correct)	- not valid while the ECU is in the Operational State	x4	Fx	xx	7F	27	xx	xx	xx	22	xx	
	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this Mode (message)	x4	Fx	xx	7F	27	xx	xx	xx	11	xx	
No Stored Codes Logging Mode	General Response [7F] of: 22 (Conditions Not Correct)	- not valid while the ECU is in the NSCL Mode	x4	Fx	xx	7F	27	xx	xx	xx	22	xx	
	Report Security Access [67]	- returns the Seed Code or the Security Access Status	x4	Fx	xx	67	27	xx	xx	xx	xx	xx	xx
DIAGNOSTIC STATE	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this Mode (message)	x4	Fx	xx	7F	27	xx	xx	xx	11	xx	
	General Response [7F] of: 12 (Subfunction Not Supported)	- invalid Security Sub-mode (if byte 5 not equal to 01 or 02)	x4	Fx	xx	7F	27	xx	xx	xx	12	xx	
Diagnostic Test Mode	General Response [7F] of: 22 (Conditions Not Correct)	- not valid, the ECU is currently executing a Diagnostic Test	x4	Fx	xx	7F	27	xx	xx	xx	22	xx	
	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this Mode (message)	x4	Fx	xx	7F	27	xx	xx	xx	11	xx	
Command Mode	General Response [7F] of: 22 (Conditions Not Correct)	- not valid while the ECU is executing a Diagnostic Command	x4	Fx	xx	7F	27	xx	xx	xx	22	xx	
	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this Mode (message)	x4	Fx	xx	7F	27	xx	xx	xx	11	xx	
Memory Transfer Mode	General Response [7F] of: 22 (Conditions Not Correct)	- not valid while the ECU is Transferring Data to/from Memory	x4	Fx	xx	7F	27	xx	xx	xx	22	xx	
	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this Mode (message)	x4	Fx	xx	7F	27	xx	xx	xx	11	xx	

REPORT SECURITY ACCESS [67]

ECU Response: that is used to report the ECU Security Seed Code or to indicate Security Access Status (granted, invalid key or denied).

Byte-by-Byte Message Description:

BYTE	VALUE	DEFINITION	SCP DESCRIPTION / COMMENTS	ISO-9141-FORD DESCRIPTION / COMMENTS
1	x4	Message Type	Priority/Header Type (C4)	ISO-9141-FORD: Length Type (A4)
2	Fx	Target Address	Tester Address	Tester Address
3	xx	Source Address	ECU Address	ECU Address
4	67	Mode Byte	Report Security Access	Report Security Access
5	xx	Access Response	01 - returning Security Access Seed, 02 - returning Security Access Status	01 - returning Security Access Seed, 02 - returning Security Access Status
6	xx	Response Data MSB	If Access Response = 01: Byte 6 - Byte 7 - Seed Code MSB Seed Code LSB	If Access Response = 01: Byte 6 - Byte 7 - Seed Code MSB Seed Code LSB
7	xx	Response Data LSB	If Access Response = 02: Byte 6 - Byte 7 - AA - Granted CC - Invalid Key 33 - Denied 00 always	If Access Response = 02: Byte 6 - Byte 7 - AA - Granted CC - Invalid Key 33 - Denied 00 always
8	xx	Current Status	If Access Response = 01: 00 - still secured 01 - Access already Granted	If Access Response = 01: 00 - still secured 01 - Access already Granted
9	xx	Data Byte 1	If Access Response = 02: 00 Security algorithm data byte 1 for request seed response when byte 5 = \$01, value of \$00 otherwise.	If Access Response = 02: 00 Security algorithm data byte 1 for request seed response when byte 5 = \$01, value of \$00 otherwise.
10	xx	Data Byte 2	Security algorithm data byte 2 for request seed response when byte 5 = \$01, value of \$00 otherwise.	Security algorithm data byte 2 for request seed response when byte 5 = \$01, value of \$00 otherwise.
11	xx	Check Byte	CRC	Checksum

REQUEST DIAGNOSTIC ROUTINE ENTRY [31]

Tester Request: that directs the ECU to execute the Diagnostic Test specified in Byte 5 of the message.

Byte-by-Byte Message Description:

BYTE	VALUE	DEFINITION	SCP DESCRIPTION / COMMENTS	ISO-9141-FORD DESCRIPTION / COMMENTS
1	x4	Message Type	Priority/Header Type (C4)	Length Type (54)
2	xx	Target Address	ECU Address	ECU Address
3	Fx	Source Address	Tester Address	Tester Address
4	31	Mode Byte	Request Diagnostic Routine Entry (request for the ECU to execute a Test)	Request Diagnostic Routine Entry (request for the ECU to execute a Test)
5	xx	Test Number	the ID Number of the Test the ECU will execute	the ID Number of the Test the ECU will execute
6	xx	Check Byte	CRC	Checksum

Valid ECU Responses to: REQUEST DIAGNOSTIC ROUTINE ENTRY:

CURRENT ECU STATE / MODE	VALID RESPONSES	REASON FOR THE RESPONSE	RESPONSE MESSAGE DATA (Byte-by-Byte)										
			1	2	3	4	5	6	7	8	9	10	11
OPERATIONAL STATE	General Response [7F] of: 22 (Conditions Not Correct)	- Can not execute a Diagnostic Test from the Operational State	x4	Fx	xx	7F	31	xx	xx	00	22	xx	
No Stored Codes Logging Mode	General Response [7F] of: 22 (Conditions Not Correct)	- Can not execute a Diagnostic Test from the NSCL Mode	x4	Fx	xx	7F	31	xx	xx	00	22	xx	
DIAGNOSTIC STATE	General Response [7F] of: 00 (Affirmative)	- the ECU shall execute the Test specified	x4	Fx	xx	7F	31	xx	xx	00	00	xx	
		- the Test Number (byte 5) is not supported by the ECU	x4	Fx	xx	7F	31	xx	xx	00	12	xx	
		- pre-conditions for Test execution have not been satisfied	x4	Fx	xx	7F	31	xx	xx	00	22	xx	
Diagnostic Test Mode	General Response [7F] of: 22 (Conditions Not Correct)	- not valid, the ECU is currently executing a Diagnostic Test	x4	Fx	xx	7F	31	xx	xx	00	22	xx	
Command Mode	General Response [7F] of: 22 (Conditions Not Correct)	- not valid while the ECU is executing a Diagnostic Command	x4	Fx	xx	7F	31	xx	xx	00	22	xx	
Memory Transfer Mode	General Response [7F] of: 22 (Conditions Not Correct)	- not valid while the ECU is Transferring Data to/from Memory	x4	Fx	xx	7F	31	xx	xx	00	22	xx	

NOTE: Upon entering a Diagnostic Routine, an ECU shall remain in the Diagnostic Test Mode until instructed to exit via a REQUEST DIAGNOSTIC ROUTINE EXIT message or until a Diagnostic Routine Exit condition has been invoked.

REQUEST DIAGNOSTIC ROUTINE EXIT [32]

Tester Request: that directs the ECU to stop execution of the Diagnostic Test (Byte 6 - Type of Exit: \$00 - if complete, \$01 - immediately) and return to the Diagnostic State.

Byte-by-Byte Message Description:

BYTE	VALUE	DEFINITION	SCP DESCRIPTION / COMMENTS	ISO 9141-FORD DESCRIPTION / COMMENTS
1	x4	Message Type	Priority/Header Type (C4)	length/type (64)
2	xx	Target Address	ECU Address	ECU Address
3	Fx	Source Address	Tester Address	Tester Address
4	32	Mode Byte	Request Diagnostic Routine Exit	Request Diagnostic Routine Exit
5	xx	Test Number	the ID Number of the Test to be exited	the ID Number of the Test to be exited
6	xx	Type of Exit	\$00 - Exit if complete, \$01 - Exit immediately	\$00 - Exit if complete, \$01 - Exit immediately
7	xx	Check Byte	CRC	Checksum

Valid ECU Responses to: REQUEST DIAGNOSTIC ROUTINE EXIT:

CURRENT ECU STATE / MODE	VALID RESPONSES	REASON FOR THE RESPONSE	RESPONSE MESSAGE DATA (Byte-by-Byte)										
			1	2	3	4	5	6	7	8	9	10	11
OPERATIONAL STATE	General Response [7F] of: 22 (Conditions Not Correct)	- Can not execute or exit a Test in the Operational State	x4	Fx	xx	7F	32	xx	00	00	22	xx	
No Stored Codes Logging Mode	General Response [7F] of: 22 (Conditions Not Correct)	- Can not execute or exit a Diagnostic Test in the NSCL Mode	x4	Fx	xx	7F	32	xx	00	00	22	xx	
DIAGNOSTIC STATE	General Response [7F] of: 22 (Conditions Not Correct)	- the ECU can not exit a Test that it is not executing	x4	Fx	xx	7F	32	xx	00	00	22	xx	
Diagnostic Test Mode	General Response [7F] of: 00 (Affirmative)	- the ECU has exited the Test specified	x4	Fx	xx	7F	32	xx	00	00	00	xx	
		- the Test Number (byte 5) or Type of Exit (byte 6) is invalid	x4	Fx	xx	7F	32	xx	00	00	12	xx	
Command Mode	General Response [7F] of: 22 (Conditions Not Correct)	- the ECU is busy executing the test (Type of Exit = \$00)	x4	Fx	xx	7F	32	xx	00	00	21	xx	
		- not valid while the ECU is executing a Diagnostic Command	x4	Fx	xx	7F	32	xx	00	00	22	xx	
Memory Transfer Mode	General Response [7F] of: 22 (Conditions Not Correct)	- not valid while the ECU is Transferring Data to/from Memory	x4	Fx	xx	7F	32	xx	00	00	22	xx	

NOTE: Upon entering a Diagnostic Routine, an ECU shall remain in the Diagnostic Test Mode until instructed to exit via a REQUEST DIAGNOSTIC ROUTINE EXIT message or until a Diagnostic Routine Exit condition has been invoked.

REQUEST DIAGNOSTIC ROUTINE RESULTS [33] (On-Demand DTCs)

Tester Request: that directs the ECU to return the results (On-Demand DTCs) of the last Diagnostic Test executed due to a Mode 31 request.

Byte-by-Byte Message Description:

BYTE	VALUE	DEFINITION	SCP DESCRIPTION / COMMENTS	ISO-9141-FORD DESCRIPTION / COMMENTS
1	x4	Message Type	Priority/Header/Type (C4)	Length/Type (54)
2	xx	Target Address	ECU Address	ECU Address
3	Fx	Source Address	Tester Address	Tester Address
4	33	Mode Byte	Request Diagnostic Routine Results (On-Demand DTCs)	Request Diagnostic Routine Results (On-Demand DTCs)
5	xx	Test Number	Test Number for which results are requested.	Test Number for which results are requested.
6	xx	Check Byte	CRC	Checksum

Valid ECU Responses to: REQUEST DIAGNOSTIC ROUTINE RESULTS:

CURRENT ECU STATE / MODE	VALID RESPONSES	REASON FOR THE RESPONSE	RESPONSE MESSAGE DATA (Byte-by-Byte)										
			1	2	3	4	5	6	7	8	9	10	11
OPERATIONAL STATE	General Response [7F] of: 22 (Conditions Not Correct)	- must be in Diagnostic State to provide Diagnostic Test results	x4	Fx	xx	7F	33	xx	00	00	22	xx	
No Stored Codes Logging Mode	General Response [7F] of: 22 (Conditions Not Correct)	- must be in Diagnostic State to provide Diagnostic Test results	x4	Fx	xx	7F	33	xx	00	00	22	xx	
DIAGNOSTIC STATE	Report Diagnostic Routine Results [73]	- returns up to 3 On-Demand DTCs per message logged during the last Tester-invoked test executed by the ECU	x4	Fx	xx	73	xx	xx	xx	xx	xx	xx	xx
		- the Test Number specified is NOT supported	x4	Fx	xx	7F	33	xx	00	00	12	xx	
		- The test is supported and the test number does NOT match most recent test ran or results requested prior to running any test											
		- not valid while the ECU is executing a Diagnostic Test	x4	Fx	xx	7F	33	xx	00	00	22	xx	
Diagnostic Test Mode	General Response [7F] of: 22 (Conditions Not Correct)	- not valid while the ECU is executing a Diagnostic Command	x4	Fx	xx	7F	33	xx	00	00	22	xx	
Command Mode	General Response [7F] of: 22 (Conditions Not Correct)	- not valid while the ECU is executing a Diagnostic Command	x4	Fx	xx	7F	33	xx	00	00	22	xx	
Memory Transfer Mode	General Response [7F] of: 22 (Conditions Not Correct)	- not valid while the ECU is Transferring Data to/from Memory	x4	Fx	xx	7F	33	xx	00	00	22	xx	

REPORT DIAGNOSTIC ROUTINE RESULTS [73]

ECU Response: that is used to report the results of the last Diagnostic Test executed which was invoked via a Mode 31 Request. (NOTE: The ECU will return consecutive responses containing up to 3 On-Demand DTCs per message until all DTCs have been reported).

Byte-by-Byte Message Description:

BYTE	VALUE	DEFINITION	SCP DESCRIPTION / COMMENTS	ISO-9141-FORD DESCRIPTION / COMMENTS
1	x4	Message Type	Priority/Header/Type (C4)	Length/Type (A4)
2	Fx	Target Address	Tester Address	Tester Address
3	xx	Source Address	ECU Address	ECU Address
4	73	Mode Byte	Report Diagnostic Routine Results	Report Diagnostic Routine Results
5	xx	MSB of Stored Code	\$00 if no On-Demand DTCs were logged by the ECU	\$00 if no On-Demand DTCs were logged by the ECU
6	xx	LSB of Stored Code	\$00 if no On-Demand DTCs were logged by the ECU	\$00 if no On-Demand DTCs were logged by the ECU
7	xx	MSB of Stored Code	\$00 if less than 2 DTCs transferred for this response	\$00 if less than 2 DTCs transferred for this response
8	xx	LSB of Stored Code	\$00 if less than 2 DTCs transferred for this response	\$00 if less than 2 DTCs transferred for this response
9	xx	MSB of Stored Code	\$00 if less than 3 DTCs transferred for this response	\$00 if less than 3 DTCs transferred for this response
10	xx	LSB of Stored Code	\$00 if less than 3 DTCs transferred for this response	\$00 if less than 3 DTCs transferred for this response
11	xx	Check Byte	CRC	Checksum

REQUEST DOWNLOAD ROUTINE ENTRY [34]

Tester Request: that directs the ECU to prepare to receive the number of data bytes designated in bytes 6 & 7 and save them beginning at the ECU Physical Address specified in bytes 8, 9 & 10. Refer to Section 11 for use.

Byte-by-Byte Message Description:

BYTE	VALUE	DEFINITION	SCP DESCRIPTION / COMMENTS	ISO-9141-FORD DESCRIPTION / COMMENTS
1	x4	Message Type	Priority/Header Type (C4)	Length Type (A4)
2	xx	Target Address	ECU Address	ECU Address
3	xx	Source Address	Tester Address	Tester Address
4	34	Mode Byte	Request Download Routine Entry	Request Download Routine Entry
5	xx	Block Transfer Format	specifies the format of data to be transferred: 80 - No Handshake 81 - Handshake, No Echo	specifies the format of data to be transferred: 80 - No Handshake 81 - Handshake, No Echo
6	xx	Byte Count - MSB	total number of bytes to be transferred (includes checksum bytes) - MSB	total number of bytes to be transferred - MSB
7	xx	Byte Count - LSB	total number of bytes to be transferred (includes checksum bytes) - LSB	total number of bytes to be transferred - LSB
8	xx	ECU Address - MSB	starting ECU Physical Address to which the data will be downloaded - MSB	starting ECU Physical Address to which the data will be downloaded - MSB
9	xx	ECU Address - Byte 2	starting ECU Physical Address to which the data will be downloaded - Byte 2	starting ECU Physical Address to which the data will be downloaded - Byte 2
10	xx	ECU Address - LSB	starting ECU Physical Address to which the data will be downloaded - LSB	starting ECU Physical Address to which the data will be downloaded - LSB
11	xx	Check Byte	CRC	Checksum

NOTE: The ECU Address Range may be extended to four Bytes by specifying the MSB in Command \$8032 - 32-bit Addressing Command, prior to sending this message.

Valid ECU Responses to: REQUEST DOWNLOAD ROUTINE ENTRY:

CURRENT ECU STATE / MODE	VALID RESPONSE	REASON FOR THE RESPONSE	RESPONSE MESSAGE DATA (Byte-by-Byte)										
OPERATIONAL STATE	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this message (Mode)	x4	Fx	xx	7F	34	xx	xx	xx	11	xx	
	22 (Conditions Not Correct)	- not valid while the ECU is in Operational State	x4	Fx	xx	7F	34	xx	xx	xx	22	xx	
No Stored Codes Logging Mode	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this message (Mode)	x4	Fx	xx	7F	34	xx	xx	xx	11	xx	
	22 (Conditions Not Correct)	- not valid while the ECU is in Operational State	x4	Fx	xx	7F	34	xx	xx	xx	22	xx	
DIAGNOSTIC STATE	General Response [7F] of: 00 (Affirmative)	- the ECU is ready to receive data	x4	Fx	xx	7F	34	xx	xx	xx	00	xx	
	11 (Mode Not Supported)	- the ECU does not support this message (Mode)	x4	Fx	xx	7F	34	xx	xx	xx	11	xx	
	12 (Subfunction Not Supported)	- Invalid: Block Transfer Format, Count or ECU Physical Address	x4	Fx	xx	7F	34	xx	xx	xx	12	xx	
Diagnostic Test Mode	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this message (Mode)	x4	Fx	xx	7F	34	xx	xx	xx	11	xx	
	22 (Conditions Not Correct)	- not valid while the ECU is executing a Diagnostic Test	x4	Fx	xx	7F	34	xx	xx	xx	22	xx	
Command Mode	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this message (Mode)	x4	Fx	xx	7F	34	xx	xx	xx	11	xx	
	22 (Conditions Not Correct)	- not valid while the ECU is executing a Diagnostic Command	x4	Fx	xx	7F	34	xx	xx	xx	22	xx	
Memory Transfer Mode	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this message (Mode)	x4	Fx	xx	7F	34	xx	xx	xx	11	xx	
	22 (Conditions Not Correct)	- not valid while the ECU is Transferring Data to/from Memory	x4	Fx	xx	7F	34	xx	xx	xx	22	xx	

REQUEST UPLOAD ROUTINE ENTRY [35]

Tester Request: that directs the ECU to prepare to transmit the number of data bytes designated in bytes 6 & 7 beginning at the ECU Physical Address specified in bytes 8, 9 & 10.

Byte-by-Byte Message Description:

BYTE	VALUE	DEFINITION	SCP DESCRIPTION / COMMENTS	ISO-9141-FORD DESCRIPTION / COMMENTS
1	x4	Message Type	Priority/Header/Type (C4)	Length/Type (A4)
2	xx	Target Address	ECU Address	ECU Address
3	Fx	Source Address	Tester Address	Tester Address
4	35	Mode Byte	Request Upload Routine Entry	Request Upload Routine Entry
5	xx	Block Transfer Format	Request Upload Routine Entry specifies the format of data to be transferred	80 - No Handshake 81 - Handshake, No Echo
6	xx	Byte Count - MSB	total number of bytes to be transferred - MSB	total number of bytes to be transferred - MSB
7	xx	Byte Count - LSB	total number of bytes to be transferred - LSB	total number of bytes to be transferred - LSB
8	xx	ECU Address - MSB	starting ECU Physical Address from which the data will be uploaded - MSB	starting ECU Physical Address from which the data will be uploaded - MSB
9	xx	ECU Address - Byte 2	starting ECU Physical Address from which the data will be uploaded - Byte 2	starting ECU Physical Address from which the data will be uploaded - Byte 2
10	xx	ECU Address - LSB	starting ECU Physical Address from which the data will be uploaded - LSB	starting ECU Physical Address from which the data will be uploaded - LSB
11	xx	Check Byte	CRC	Checksum

NOTE: The ECU Address Range may be extended to four Bytes by specifying the MSB in Command \$8032 - 32-bit Addressing Command, prior to sending this message

Valid ECU Responses to: REQUEST UPLOAD ROUTINE ENTRY:

CURRENT ECU STATE / MODE	VALID RESPONSES	REASON FOR THE RESPONSE	RESPONSE MESSAGE DATA (Byte-by-Byte)
OPERATIONAL STATE	General Response [7F] of: 11 (Mode Not Supported) 22 (Conditions Not Correct)	- the ECU does not support this message (Mode) - not valid while the ECU is in the Operational State	1 x4 2 Fx 3 xx 4 7F 5 35 6 xx 7 xx 8 xx 9 11 10 xx 11 xx
No Stored Codes Logging Mode	General Response [7F] of: 11 (Mode Not Supported) 22 (Conditions Not Correct)	- the ECU does not support this message (Mode) - not valid while the ECU is in the Operational State	1 x4 2 Fx 3 xx 4 7F 5 35 6 xx 7 xx 8 xx 9 11 10 xx 11 xx
DIAGNOSTIC STATE	General Response [7F] of: 00 (Affirmative) 11 (Mode Not Supported) 12 (Subfunction Not Supported)	- OK to transfer data from the ECU - the ECU does not support this message (Mode) - Invalid Format, Count or Starting Address	1 x4 2 Fx 3 xx 4 7F 5 35 6 xx 7 xx 8 xx 9 11 10 xx 11 xx
Diagnostic Test Mode	General Response [7F] of: 11 (Mode Not Supported) 22 (Conditions Not Correct)	- the ECU does not support this message (Mode) - not valid while the ECU is executing a Diagnostic Test	1 x4 2 Fx 3 xx 4 7F 5 35 6 xx 7 xx 8 xx 9 11 10 xx 11 xx
Command Mode	General Response [7F] of: 11 (Mode Not Supported) 22 (Conditions Not Correct)	- the ECU does not support this message (Mode) - not valid while the ECU is executing a Diagnostic Command	1 x4 2 Fx 3 xx 4 7F 5 35 6 xx 7 xx 8 xx 9 11 10 xx 11 xx
Memory Transfer Mode	General Response [7F] of: 11 (Mode Not Supported) 22 (Conditions Not Correct)	- the ECU does not support this message (Mode) - not valid while the ECU is Transferring Data to/from Memory	1 x4 2 Fx 3 xx 4 7F 5 35 6 xx 7 xx 8 xx 9 11 10 xx 11 xx

BLOCK DATA TRANSFER MESSAGE [36]

Tester Message: that contains data to be Downloaded from the Tester to the ECU

<OR>

ECU Message: that contains data to be Uploaded from the ECU to the Tester

Byte-by-Byte Message Description:

BYTE	VALUE	DEFINITION	SCP DESCRIPTION / COMMENTS	ISO-9141-FORD DESCRIPTION / COMMENTS
1	x4	Message Type	Priority/Header/Type (C4)	Length/Type (A4)
2	xx	Target Address	ECU Address if Downloading, Tester Address if Uploading	ECU Address if Downloading, Tester Address if Uploading
3	xx	Source Address	Tester Address if Downloading, ECU Address if Uploading	Tester Address if Downloading, ECU Address if Uploading
4	36	Mode Byte	Block Data Transfer Message	Block Data Transfer Message
5	xx	Data Byte #1	1st byte of data	1st byte of data
6	xx	Data Byte #2	2nd byte of data or \$00 if less than 2 bytes will be transferred	2nd byte of data or \$00 if less than 2 bytes will be transferred
7	xx	Data Byte #3	3rd byte of data or \$00 if less than 3 bytes will be transferred	3rd byte of data or \$00 if less than 3 bytes will be transferred
8	xx	Data Byte #4	4th byte of data or \$00 if less than 4 bytes will be transferred	4th byte of data or \$00 if less than 4 bytes will be transferred
9	xx	Data Byte #5	5th byte of data or \$00 if less than 5 bytes will be transferred	5th byte of data or \$00 if less than 5 bytes will be transferred
10	xx	Data Byte #6	6th byte of data or \$00 if less than 6 bytes will be transferred	6th byte of data or \$00 if less than 6 bytes will be transferred
11	xx	Check Byte	ORC	Checksum

Valid ECU Responses to: BLOCK DATA TRANSFER MESSAGE:

CURRENT ECU STATE / MODE	VALID RESPONSES	REASON FOR THE RESPONSE	RESPONSE MESSAGE DATA (Byte-by-Byte)										
OPERATIONAL STATE	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this message (Mode)	x4	Fx	xx	7F	36	xx	xx	xx	11	xx	
	22 (Conditions Not Correct)	- must be in Memory Transfer Mode to transfer data	x4	Fx	xx	7F	36	xx	xx	xx	22	xx	
No Stored Codes Logging Mode	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this message (Mode)	x4	Fx	xx	7F	36	xx	xx	xx	11	xx	
	22 (Conditions Not Correct)	- must be in Memory Transfer Mode to transfer data	x4	Fx	xx	7F	36	xx	xx	xx	22	xx	
DIAGNOSTIC STATE	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this message (Mode)	x4	Fx	xx	7F	36	xx	xx	xx	11	xx	
	22 (Conditions Not Correct)	- must be in Memory Transfer Mode to transfer data	x4	Fx	xx	7F	36	xx	xx	xx	22	xx	
Diagnostic Test Mode	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this message (Mode)	x4	Fx	xx	7F	36	xx	xx	xx	11	xx	
	22 (Conditions Not Correct)	- the ECU does not support this message (Mode)	x4	Fx	xx	7F	36	xx	xx	xx	22	xx	
Command Mode	General Response [7F] of: 11 (Mode Not Supported)	- must be in Memory Transfer Mode to transfer data	x4	Fx	xx	7F	36	xx	xx	xx	11	xx	
	22 (Conditions Not Correct)	- the ECU does not support this message (Mode)	x4	Fx	xx	7F	36	xx	xx	xx	22	xx	
Memory Transfer Mode	General Response [7F] of: 00 (Affirmative)	- Affirmative for data received. Handshake for format 80 responds to each block transferred. Non-handshake for format 81 responds after all blocks have been transferred	x4	Fx	xx	7F	36	xx	xx	xx	00	xx	
	12 (Invalid data)	- Invalid data was present in the block transfer message	x4	Fx	xx	7F	36	xx	xx	xx	12	xx	
	21 (Busy)	- Module is busy	x4	Fx	xx	7F	36	xx	xx	xx	21	xx	

REQUEST BLOCK TRANSFER EXIT [37]

Tester Request: that directs the ECU to halt the Downloading or Uploading sequence.

Byte-by-Byte Message Description:

BYTE	VALUE	DEFINITION	SCP DESCRIPTION / COMMENTS	ISO-9141-FORD DESCRIPTION / COMMENTS
1	x4	Message Type	Priority/Header/Type (C4)	length/type (54)
2	xx	Target Address	ECU Address	ECU Address
3	xx	Source Address	Tester Address	Tester Address
4	37	Mode Byte	Request Block Transfer Exit	Request Block Transfer Exit
5	xx	Block Transfer Format	Specifies the format of the data transfer	80 - No Handshake 81 - Handshake, No Echo
6	xx	Check Byte	CRC	Checksum

Valid ECU Responses to: REQUEST BLOCK TRANSFER EXIT:

CURRENT ECU STATE / MODE	VALID RESPONSES	REASON FOR THE RESPONSE	RESPONSE MESSAGE DATA (Byte-by-Byte)										
			1	2	3	4	5	6	7	8	9	10	11
OPERATIONAL STATE	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this message (Mode)	Fx	Fx	xx	7F	37	xx	00	00	11	xx	
		- must be in Memory Transfer Mode to exit	Fx	Fx	xx	7F	37	xx	00	00	22	xx	
No Stored Codes Logging Mode	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this message (Mode)	Fx	Fx	xx	7F	37	xx	00	00	11	xx	
		- must be in Memory Transfer Mode to exit	Fx	Fx	xx	7F	37	xx	00	00	22	xx	
DIAGNOSTIC STATE	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this message (Mode)	Fx	Fx	xx	7F	37	xx	00	00	11	xx	
		- must be in Memory Transfer Mode to exit	Fx	Fx	xx	7F	37	xx	00	00	22	xx	
Diagnostic Test Mode	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this message (Mode)	Fx	Fx	xx	7F	37	xx	00	00	11	xx	
		- must be in Memory Transfer Mode to exit	Fx	Fx	xx	7F	37	xx	00	00	22	xx	
Command Mode	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this message (Mode)	Fx	Fx	xx	7F	37	xx	00	00	11	xx	
		- must be in Memory Transfer Mode to exit	Fx	Fx	xx	7F	37	xx	00	00	22	xx	
Memory Transfer Mode	General Response [7F] of: 00 (Affirmative)	- exiting Memory Transfer Mode	x4	Fx	xx	7F	37	xx	00	00	12	xx	
		- invalid Block Transfer Format (byte 5)	x4	Fx	xx	7F	37	xx	00	00	12	xx	
		21 (Busy)	x4	Fx	xx	7F	37	xx	00	00	21	xx	

REQUEST WRITE MEMORY BLOCK [3B]

Tester Request: that directs the ECU to write the data provided (bytes 6-10) into its memory beginning at the location specified by the logical Block Number (byte 5).

Byte-by-Byte Message Description:

BYTE	VALUE	DEFINITION	SCP DESCRIPTION / COMMENTS	ISO-9141-FORD DESCRIPTION / COMMENTS
1	x4	Message Type	Priority/Header Type (C4)	Length Type (A4)
2	xx	Target Address	ECU Address	ECU Address
3	xx	Source Address	Tester Address	Tester Address
4	3B	Mode Byte	Request Write Memory Block	Request Write Memory Block
5	xx	Block Number	pointer to an ECU physical address where the data will be written (logical)	pointer to an ECU physical address where the data will be written (logical)
6	xx	Data Byte 1	1st byte of data to be written	1st byte of data to be written
7	xx	Data Byte 2	2nd byte of data to be written or padded with 00	2nd byte of data to be written or padded with 00
8	xx	Data Byte 3	3rd byte of data to be written or padded with 00	3rd byte of data to be written or padded with 00
9	xx	Data Byte 4	4th byte of data to be written or padded with 00	4th byte of data to be written or padded with 00
10	xx	Data Byte 5	5th byte of data to be written or padded with 00	5th byte of data to be written or padded with 00
11	xx	Check Byte	CRC	Checksum

Valid ECU Responses to: REQUEST WRITE MEMORY BLOCK:

VALID RESPONSES			REASON FOR THE RESPONSE			RESPONSE MESSAGE DATA (Byte-by-Byte)										
CURRENT ECU STATE / MODE	VALID RESPONSES	REASON FOR THE RESPONSE	1	2	3	4	5	6	7	8	9	10	11			
OPERATIONAL STATE	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this message (Mode)	x4	Fx	xx	7F	3B	xx	xx	xx	xx	11	xx			
	22 (Conditions Not Correct)	- must be in Diagnostic State	x4	Fx	xx	7F	3B	xx	xx	xx	xx	22	xx			
No Stored Codes Logging Mode	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this message (Mode)	x4	Fx	xx	7F	3B	xx	xx	xx	xx	11	xx			
	22 (Conditions Not Correct)	- must be in Diagnostic State	x4	Fx	xx	7F	3B	xx	xx	xx	xx	22	xx			
DIAGNOSTIC STATE	General Response [7F] of: 00 (Affirmative)	- correct data has been received and will be written to memory	x4	Fx	xx	7F	3B	xx	xx	xx	xx	00	xx			
	11 (Mode Not Supported)	- the ECU does not support this message (Mode)	x4	Fx	xx	7F	3B	xx	xx	xx	xx	11	xx			
	12 (Subfunction Not Supported)	- the Block Number or one or more data bytes is not correct	x4	Fx	xx	7F	3B	xx	xx	xx	xx	12	xx			
	21 (Busy)	- the ECU is currently Busy with normal operation - try again	x4	Fx	xx	7F	3B	xx	xx	xx	xx	21	xx			
Diagnostic Test Mode	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this message (Mode)	x4	Fx	xx	7F	3B	xx	xx	xx	xx	11	xx			
	22 (Conditions Not Correct)	- must be in Diagnostic State	x4	Fx	xx	7F	3B	xx	xx	xx	xx	22	xx			
Command Mode	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this message (Mode)	x4	Fx	xx	7F	3B	xx	xx	xx	xx	11	xx			
	22 (Conditions Not Correct)	- must be in Diagnostic State	x4	Fx	xx	7F	3B	xx	xx	xx	xx	22	xx			
Memory Transfer Mode	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this message (Mode)	x4	Fx	xx	7F	3B	xx	xx	xx	xx	11	xx			
	22 (Conditions Not Correct)	- must be in Diagnostic State	x4	Fx	xx	7F	3B	xx	xx	xx	xx	22	xx			

REQUEST READ MEMORY BLOCK [3C]

Tester Request: that directs the ECU to return the data from its memory beginning at the location specified by the logical Block Number (byte 5).

Byte-by-Byte Message Description:

BYTE	VALUE	DEFINITION	SCP DESCRIPTION / COMMENTS	ISO-9141-FORD DESCRIPTION / COMMENTS
1	x4	Message Type	Priority/Header/Type (C4)	Length/Type (54)
2	xx	Target Address	ECU Address	ECU Address
3	Fx	Source Address	Tester Address	Tester Address
4	3C	Mode Byte	Request Read Memory Block	Request Read Memory Block
5	xx	Block Number	pointer to an ECU physical address from where the data will be read (logical)	pointer to an ECU physical address from where the data will be read (logical)
6	xx	Check Byte	CRC	Checksum

Valid ECU Responses to: REQUEST READ MEMORY BLOCK:

CURRENT ECU STATE / MODE	INVALID RESPONSES	REASON FOR THE RESPONSE	RESPONSE MESSAGE DATA (Byte-by-Byte)										
			1	2	3	4	5	6	7	8	9	10	11
OPERATIONAL STATE	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this message (Mode)	x4	Fx	xx	7F	3C	xx	00	00	11	xx	
	22 (Conditions not correct)	- Must be in diagnostic state	x4	Fx	xx	7F	3C	xx	00	00	22	xx	
No Stored Codes Logging Mode	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this message (Mode)	x4	Fx	xx	7F	3C	xx	00	00	11	xx	
	22 (Conditions not correct)	- Must be in diagnostic state	x4	Fx	xx	7F	3C	xx	00	00	22	xx	
DIAGNOSTIC STATE	Report Contents of Memory Block [7C]	- to report 5 bytes of data from memory beginning at the location pointed to by the Block Number	x4	Fx	xx	7C	xx	xx	xx	xx	xx	xx	xx
	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this message (Mode)	x4	Fx	xx	7F	3C	xx	00	00	11	xx	
Diagnostic Test Mode	12 (Subfunction Not Supported)	- the Block Number or one or more data bytes is not correct	x4	Fx	xx	7F	3C	xx	00	00	12	xx	
	General Response [7F] of: 11 (Mode Not Supported)	- must be in Diagnostic State	x4	Fx	xx	7F	3C	xx	00	00	11	xx	
Command Mode	22 (Conditions Not Correct)	- must be in Diagnostic State	x4	Fx	xx	7F	3C	xx	00	00	22	xx	
	General Response [7F] of: 11 (Mode Not Supported)	- must be in Diagnostic State	x4	Fx	xx	7F	3C	xx	00	00	11	xx	
Memory Transfer Mode	General Response [7F] of: 22 (Conditions Not Correct)	- must be in Diagnostic State	x4	Fx	xx	7F	3C	xx	00	00	22	xx	
	General Response [7F] of: 11 (Mode Not Supported)	- must be in Diagnostic State	x4	Fx	xx	7F	3C	xx	00	00	11	xx	
	General Response [7F] of: 22 (Conditions Not Correct)	- must be in Diagnostic State	x4	Fx	xx	7F	3C	xx	00	00	22	xx	

REPORT CONTENTS OF MEMORY BLOCK [7C]

ECU Response: that is used to report up to 5 bytes of data from memory beginning at the location pointed to by the Block Number.

Byte-by-Byte Message Description:

BYTE	VALUE	DEFINITION	SCP DESCRIPTION / COMMENTS	ISO-9141-FORD DESCRIPTION / COMMENTS
1	x4	Message Type	Priority/Header/Type (C4)	Length/Type (A4)
2	Fx	Target Address	Tester Address	Tester Address
3	xx	Source Address	ECU Address	ECU Address
4	7C	Mode Byte	Report Contents of Memory Block	Report Contents of Memory Block
5	xx	Block Number	pointer to an ECU physical address from which data will be retrieved (logical)	pointer to an ECU physical address from which data will be retrieved (logical)
6	xx	Data Byte 1	1st Byte of data to be read	1st Byte of data to be read
7	xx	Data Byte 2	2nd Byte of data to be read or padded with 00	2nd Byte of data to be read or padded with 00
8	xx	Data Byte 3	3rd Byte of data to be read or padded with 00	3rd Byte of data to be read or padded with 00
9	xx	Data Byte 4	4th Byte of data to be read or padded with 00	4th Byte of data to be read or padded with 00
10	xx	Data Byte 5	5th Byte of data to be read or padded with 00	5th Byte of data to be read or padded with 00
11	xx	Check Byte	CRC	Checksum

TESTER PRESENT [3F] (PCM Only - NO DEVIATIONS SHALL BE GRANTED TO NON-PCM MODULES)

Tester Request: issued periodically to inform the PCM that the Tester is still powered-on and attached to the Data Link Connector.

Byte-by-Byte Message Description:

BYTE	VALUE	DEFINITION	SCP DESCRIPTION / COMMENTS	ISO-9141-FORD DESCRIPTION / COMMENTS
1	x4	Message Type	Priority/Header/Type (C4)	Length/Type (44)
2	xx	Target Address	ECU Address	ECU Address
3	Fx	Source Address	Tester Address	Tester Address
4	3F	Mode Byte	Tester Present	Tester Present
5	xx	Check Byte	CRC	ISO-9141-FORD: Checksum

Valid ECU Responses to: TESTER PRESENT:

CURRENT ECU STATE / MODE	VALID RESPONSES	REASON FOR THE RESPONSE	RESPONSE MESSAGE DATA (Byte-by-Byte)										
			1	2	3	4	5	6	7	8	9	10	11
OPERATIONAL STATE	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this message (Mode)	x4	Fx	xx	7F	3F	00	00	00	11	xx	
	00 (Affirmative - PCM Only)	- acknowledge presence of Tester	x4	Fx	xx	7F	3F	00	00	00	00	xx	
No Stored Codes Logging Mode	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this message (Mode)	x4	Fx	xx	7F	3F	00	00	00	11	xx	
	00 (Affirmative - PCM Only)	- acknowledge presence of Tester	x4	Fx	xx	7F	3F	00	00	00	00	xx	
DIAGNOSTIC STATE	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this message (Mode)	x4	Fx	xx	7F	3F	00	00	00	11	xx	
	00 (Affirmative - PCM Only)	- acknowledge presence of Tester	x4	Fx	xx	7F	3F	00	00	00	00	xx	
Diagnostic Test Mode	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this message (Mode)	x4	Fx	xx	7F	3F	00	00	00	11	xx	
	00 (Affirmative - PCM Only)	- acknowledge presence of Tester	x4	Fx	xx	7F	3F	00	00	00	00	xx	
Command Mode	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this message (Mode)	x4	Fx	xx	7F	3F	00	00	00	11	xx	
	00 (Affirmative - PCM Only)	- acknowledge presence of Tester	x4	Fx	xx	7F	3F	00	00	00	00	xx	
Memory Transfer Mode	General Response [7F] of: 11 (Mode Not Supported)	- the ECU does not support this message (Mode)	x4	Fx	xx	7F	3F	00	00	00	11	xx	
	00 (Affirmative - PCM Only)	- acknowledge presence of Tester	x4	Fx	xx	7F	3F	00	00	00	00	xx	

REQUEST NO STORED CODES LOGGING [B0] (SCP Only)

Tester Request: that directs the ECU to stop logging Continuous DTCs (enter a No Stored Codes Logging - NSCL operating Mode)

Byte-by-Byte Message Description:

BYTE	VALUE	DEFINITION	SCP DESCRIPTION / COMMENTS	ISO-9141-FORD DESCRIPTION / COMMENTS
1	C1	Message Type	Priority/Header/Type (C1) - Functional Broadcast Message	N/A
2	5A	Target Functional Address	Non-Legislated Diagnostic Functional Address	N/A
3	Fx	Source Address	Tester Address	N/A
4	B0	Mode Byte	Request No Stored Codes Logging	N/A
5	xx	Response Byte	00 - no response from the ECU is necessary	N/A
6	xx	Check Byte	CRC	N/A

Valid ECU Responses to: REQUEST NO STORED CODES LOGGING:

CURRENT ECU STATE / MODE	VALID RESPONSES	REASON FOR THE RESPONSE	RESPONSE MESSAGE DATA (Byte-by-Byte)										
			1	2	3	4	5	6	7	8	9	10	11
OPERATIONAL STATE	NONE REQUIRED (only Hardware Acknowledged)	- N/A	NONE REQUIRED										
No Stored Codes Logging Mode	NONE REQUIRED (only Hardware Acknowledged)	- N/A	NONE REQUIRED										
DIAGNOSTIC STATE	NONE REQUIRED (only Hardware Acknowledged)	- N/A	NONE REQUIRED										
Diagnostic Test Mode	NONE REQUIRED (only Hardware Acknowledged)	- N/A	NONE REQUIRED										
Command Mode	NONE REQUIRED (only Hardware Acknowledged)	- N/A	NONE REQUIRED										
Memory Transfer Mode	NONE REQUIRED (only Hardware Acknowledged)	- N/A	NONE REQUIRED										

DIAGNOSTIC COMMAND [B1]

Tester Request: that directs the ECU to execute the Command specified in bytes 5 and 6 of the message.

Byte-by-Byte Message Description:

BYTE	VALUE	DEFINITION	SCP DESCRIPTION / COMMENTS	ISO 9141-FORD DESCRIPTION / COMMENTS
1	x4	Message Type	Priority/Header/Type (C4)	Length/Type (x4)
2	xx	Target Address	ECU Address	ECU Address
3	Fx	Source Address	Tester Address	Tester Address
4	B1	Mode Byte	Diagnostic Command	Diagnostic Command
5	xx	Diagnostic Command MSB	MSB of the Diagnostic Command Number	MSB of the Diagnostic Command Number
6	xx	Diagnostic Command LSB	LSB of the Diagnostic Command Number	LSB of the Diagnostic Command Number
7	xx	Command data byte 1	To be used as defined in MRDB and SSDS	To be used as defined in MRDB and SSDS
8	xx	Command data byte 2	To be used as defined in MRDB and SSDS	To be used as defined in MRDB and SSDS
9	xx	Command data byte 3	To be used as defined in MRDB and SSDS	To be used as defined in MRDB and SSDS
10	xx	Command data byte 4	To be used as defined in MRDB and SSDS	To be used as defined in MRDB and SSDS
11	xx	Check Byte	CRC	Checksum

Valid ECU Responses to: DIAGNOSTIC COMMAND REQUEST:

CURRENT ECU STATE / MODE	VALID RESPONSES	REASON FOR THE RESPONSE	RESPONSE MESSAGE DATA (Byte-by-Byte)										
			1	2	3	4	5	6	7	8	9	10	11
OPERATIONAL STATE	General Response [7F] of: 22 (Conditions Not Correct)	- not valid in Operational State, must be in Diagnostic State	x4	Fx	xx	7F	B1	xx	xx	xx	22	xx	
	11 (Mode Not Supported)	- The ECU does Not Support This Mode	x4	Fx	xx	7F	B1	xx	xx	xx	11	xx	
No Stored Codes Logging Mode	General Response [7F] of: 22 (Conditions Not Correct)	- not valid in the NSCL Mode, must be in the Diagnostic State	x4	Fx	xx	7F	B1	xx	xx	xx	22	xx	
	11 (Mode Not Supported)	- The ECU does Not Support This Mode	x4	Fx	xx	7F	B1	xx	xx	xx	11	xx	
DIAGNOSTIC STATE	General Response [7F] of: 00 (Affirmative)	- the ECU will perform the Command instruction received	x4	Fx	xx	7F	B1	xx	xx	xx	00	xx	
	11 (Mode Not Supported)	- The ECU does Not Support This Mode	x4	Fx	xx	7F	B1	xx	xx	xx	11	xx	
	12 (Subfunction Not Supported)	- Invalid Command Number (Bytes 5 & 6) or Data (Bytes 7 -10)	x4	Fx	xx	7F	B1	xx	xx	xx	12	xx	
	22 (Conditions Not Correct)	- Setup conditions not correct	x4	Fx	xx	7F	B1	xx	xx	xx	22	xx	
Diagnostic Test Mode	General Response [7F] of: 22 (Conditions Not Correct)	- not valid while the ECU is executing a Diagnostic Test	x4	Fx	xx	7F	B1	xx	xx	xx	22	xx	
	11 (Mode Not Supported)	- The ECU does Not Support This Mode	x4	Fx	xx	7F	B1	xx	xx	xx	11	xx	
Command Mode	General Response [7F] of: 22 (Conditions Not Correct)	- not valid, the ECU is currently executing Command	x4	Fx	xx	7F	B1	xx	xx	xx	22	xx	
	11 (Mode Not Supported)	- The ECU does Not Support This Mode	x4	Fx	xx	7F	B1	xx	xx	xx	11	xx	
Memory Transfer Mode	General Response [7F] of: 22 (Conditions Not Correct)	- not valid while the ECU is transferring data to/from Memory	x4	Fx	xx	7F	B1	xx	xx	xx	22	xx	
	11 (Mode Not Supported)	- The ECU does Not Support This Mode	x4	Fx	xx	7F	B1	xx	xx	xx	11	xx	

Appendix B GLOBAL DIAGNOSTIC SPECIFICATION (Part I) - Version 4.2 Requirements Checklist

Requirement		Section	Mandatory or Optional
1	Any deviation to this "Global Diagnostic Specification ", for any ECU Subsystem, shall only be valid if agreed upon in whole by the Worldwide Diagnostics Organization (WWDO), diagnostic application engineer.	1.1	Mandatory
2	Any deviation so granted shall be fully documented and adequately explained in the SSDS for the ECU subsystem.	1.1	Mandatory
3	Each Diagnostic SSDS for the ECU shall adhere to all of the diagnostic requirements and recommended diagnostic practices described in this document.	1.1.1	Mandatory
4	ECUs shall use the following fault probability ranking to evaluate I/O faults: 1) open circuit, 2) intermittent open circuit, 3) intermittent short to ground, 4) short to ground, 5) battery short, 6) intermittent short to battery, 7) signal fault	2.4.2	Mandatory
5	An Open Circuit is the most prevalent type of I/O fault and shall be given a higher priority consideration for diagnostic coverage.	2.4.2	Mandatory
6	ECU's shall be capable of detecting the top 80 percent (highest probability of occurrence) of the prioritized system/subsystem faults listed in the FMEA.	2.5	Mandatory
7	Of the remaining 20 percent of the faults listed in the FMEA, ECU's shall consider identifying all faults which would have a significant impact on Warranty Costs [part costs, labor costs, and Trouble Not Identified (TNI) rates].	2.5	Mandatory
8	ECU input and output fault symptoms (customer-observed) recorded and identified in a FMEA summary document shall be correlated to Diagnostic Trouble Codes (DTCs - Continuous and On-Demand) in the "DTCs" section of the SSDS.	2.6	Mandatory
9	ECU input and output fault symptoms information shall be further correlated to I/O specific Parameter Identifier (PID) and Command information in the "OVERVIEW" section of the SSDS for each ECU.	2.6	Mandatory
10	While in the Operational State, an ECU shall support continuous diagnostic trouble codes (DTC)	4.2 and Figure 4-1	Mandatory
11	While in the Operational State, an ECU shall support access to parameter identification (PID) information.	4.2 and Figure 4-1	Mandatory
12	While in the Operational State, an ECU shall support data access via direct memory reference (DMR).	4.2 and Figure 4-1	Optional
13	While in the Operational State, an ECU shall support entry into the diagnostic state.	4.2 and Figure 4-1	Mandatory
14	While in the Operational State, an ECU shall support the network diagnostic communication strategies (SCP Only).	4.2	Mandatory
15	While in the Operational State, an ECU shall support entry into the No Stored Codes Logging (NSCL) mode.	4.2 and Figure 4-1	Optional
16	While in the Operational State, an ECU shall support wake-up from a message from a Tester or a designated "wake-up" input.	4.2	Mandatory

	Requirement	Section	Mandatory or Optional
17	While in the Operational State, an ECU shall support detecting subsystem component faults.	4.2.1	Mandatory
18	While in the Operational State, an ECU shall support logging continuous DTCs.	4.2.1	Mandatory
19	While in the Operational State, an ECU shall support reporting logged Continuous DTCs over the serial communications link upon request.	4.2.1	Mandatory
20	While in the Operational State, an ECU shall support clearing logged Continuous DTCs upon receiving a request over the serial communications link	4.2.1	Mandatory
21	While in the Operational State, an ECU shall support automatic clearing of Continuous DTCs.	4.2.1	Mandatory
22	While in the Operational State, an ECU shall be capable of interpreting PID requests and reporting the requested information.	4.2.2	Mandatory
23	While in the Operational State, an ECU shall be capable of interpreting DMR requests and reporting the requested information.	4.2.3	Optional
24	An ECU shall enter the Diagnostic State upon receiving a Request Diagnostic State Entry message over the serial communications link, if all entry criteria specified in the ECUs SSDS have been satisfied.	4.2.4	Mandatory
25	An ECU in the Operational State shall enter the No Stored Codes Logging mode upon receiving a Request No Stored Codes Logging message over the SCP communications link (SCP Only).	4.2.6	Optional
26	A sleep node ECU shall begin normal operation after receiving some form of "wake-up" stimulus, either a message from a non-sleep node ECU (SCP only), a message from a Tester, or a transition on a designated "wake-up" input.	4.2.7.2	Mandatory
27	A sleep node ECU shall be able to interpret and respond appropriately to all subsequent messages, after waking-up.	4.2.7.2	Mandatory
28	A sleep node ECU shall "wake-up" upon receiving a message from a Tester.	4.2.7.2	Mandatory
29	To enter Diagnostic State from the Operational State, or from the No Stored Codes Logging Mode, an ECU shall meet all entry conditions specified in the ECU's SSDS.	5.2	Mandatory
30	If all entry conditions are satisfied, the ECU shall enter the diagnostic state upon receiving a Request Diagnostic State Entry message.	5.2	Mandatory
31	If the ECU receives a Request Diagnostic State Entry message and entry conditions are not met, the ECU shall not enter the Diagnostic State the ECU shall return a General Response message with the response code set to \$22- conditions not correct.	5.2	Mandatory
32	Upon entering the Diagnostic State, all ECU controlled outputs shall be made inactive; unless specified in the ECU's SSDS.	5.2.1	Mandatory
33	Upon entering the diagnostic state, all ECU inputs shall be isolated from directly controlling related ECU outputs.	5.2.1	Mandatory
34	While in the Diagnostic State, the ECU shall maintain a 5.0-second (-0.0, +1.0) timer that shall be reset each time it receives ANY Node-to-Node message.	5.3	Mandatory
35	While in the Diagnostic State, the ECU shall ignore any broadcast message received over the SCP serial communications link.	5.3	Mandatory
36	While in the diagnostic state, the ECU shall support diagnostic tests.	5.3	Mandatory
37	While in the diagnostic state, the ECU shall support diagnostic commands.	5.3	Optional
38	While in the diagnostic state, the ECU shall support continuous DTC requirements.	5.3	Mandatory
39	While in the diagnostic state, the ECU shall support on demand DTC requirements.	5.3	Mandatory
40	While in the diagnostic state, the ECU shall support PID requirements.	5.3	Mandatory

Requirement		Section	Mandatory or Optional
41	While in the diagnostic state, the ECU shall support DMR.	5.3	Optional
42	While in the diagnostic state, the ECU shall support memory transfer.	5.3	Optional
43	No Continuous DTCs shall be logged by an ECU while it is executing in the Diagnostic State; except DTC A141 and A477.	5.3	Mandatory
44	Continuous DTCs previously logged shall be maintained (i.e., not overwritten) by the ECU while it is executing in the Diagnostic State.	5.3	Mandatory
45	While in the diagnostic state, an ECU shall support reporting logged continuous DTCs over the serial communication link upon request.	5.3.3	Mandatory
46	While in the diagnostic state, an ECU shall support clearing logged continuous DTCs upon receiving a request over the serial communication link.	5.3.3	Mandatory
47	While in the diagnostic state, an ECU shall support logging On-Demand DTCs that identify subsystem failures detected during the execution of a diagnostic test.	5.3.4	Mandatory
48	While in the diagnostic state, an ECU shall support reporting On-Demand DTCs over the serial communication link upon request.	5.3.4	Mandatory
49	While in the Diagnostic State, an ECU shall be capable of interpreting PID requests and reporting the requested information.	5.3.5	Mandatory
50	While in the Diagnostic State, an ECU shall be capable of interpreting DMR requests and reporting the requested information.	5.3.6	Optional
51	While in the Diagnostic State, an ECU shall be capable of receiving downloaded data from a Tester and uploading data to a Tester.	5.3.7	Optional
52	An ECU shall exit the Diagnostic State and return to the operational state upon receiving a Request Return to Operational State message over the serial communications link.	5.4	Mandatory
53	An ECU shall exit the diagnostic state and return to the operational state if the ECU does not receive a valid Node-to-Node message within every 5.0 second (-0.0, +1.0) interval, after entering the Diagnostic State.	5.4	Mandatory
54	Upon returning to the Operational State, an ECU shall return all I/O to their normal operating states and resume normal operation.	5.4.1	Mandatory
55	Upon returning to the Operational State, an ECU shall return to normal operation within 500 milliseconds.	5.4.2	Mandatory
56	Upon returning to the Operational State, all On-Demand DTCs logged during execution of a Diagnostic Test shall be cleared.	5.4.3	Mandatory
57	All ECUs shall support an On-Demand Self Test.	6.2	Mandatory
58	An ECUs On-Demand Self Test shall test the top 80% of the faults listed in the FMEA.	6.2.1.1	Mandatory
59	Upon completion of the On-Demand Self-Test, the ECU shall return its outputs to their initial entry conditions (in existence prior to the ECU entering On-Demand Self-Test).	6.2.1.1	Mandatory
60	An ECU shall complete its On-Demand Self-Test within 30 seconds after receiving the request to perform the test.	6.2.1.2	Mandatory
61	An ECU shall omit evaluation of subsystem I/O circuits during on demand self test, if testing of a circuit would cause the 30 second on demand self test execution time limit to be exceeded.	6.2.1.3	Mandatory
62	An ECU shall omit evaluation of subsystem I/O circuits during on demand self test, if testing of a circuit would require operator interaction during the test.	6.2.1.3	Mandatory

Requirement		Section	Mandatory or Optional
63	An ECU shall omit evaluation of subsystem I/O circuits during on demand self test, if testing of a circuit could damage ECU subsystem components.	6.2.1.3	Mandatory
64	An ECU shall omit evaluation of subsystem I/O circuits during on demand self test, if testing of a circuit could cause hazardous conditions during the test.	6.2.1.3	Mandatory
65	An ECU shall omit evaluation of subsystem I/O circuits during on demand self test, if the reliability of the circuit is so good that it does not fail within the top 80% of the faults listed in the FMEA for the ECU subsystem.	6.2.1.3	Mandatory
66	During a wiggle test, upon detecting a change of state fault on any I/O line, the ECU shall log a On-Demand DTC that identifies that fault and increment the value in PID \$0201 to reflect the number of individual I/O transitions detected.	6.2.3	Optional
67	An ECU shall execute a Diagnostic Test upon receiving a Request Diagnostic Routine Entry message from a Tester and if the entry requirements are satisfied.	6.2.4	Mandatory
68	Upon executing a Diagnostic Test, an ECU shall remain in the test mode until instructed to exit.	6.2.4	Mandatory
69	An ECU shall exit a Diagnostic Test and return to the Diagnostic State if the ECU receives a Request Diagnostic Routine Exit message over the serial communication link.	6.2.6.1	Mandatory
70	An ECU shall exit a Diagnostic Test and return to the Diagnostic State if an Exit condition, as specified in the ECUs SSDS, that causes the ECU to return to the Diagnostic State, is met.	6.2.6.1	Mandatory
71	An ECU shall exit a Diagnostic Test and return to the Operational State if the ECU receives a Request Return to Operational State message over the serial communications link	6.2.6.2	Mandatory
72	An ECU shall exit a Diagnostic Test and return to the Operational State if the ECU does not continue to receive a valid Node-to-Node message within every 5.0-second (-0.0, +1.0) interval while executing a Diagnostic Test.	6.2.6.2	Mandatory
73	An ECU shall exit a Diagnostic Test and return to the Operational State if an Exit condition, as specified in the ECUs SSDS, that causes the ECU to return to the Operational State, is met.	6.2.6.2	Mandatory
74	While in Diagnostic State, an ECU shall be capable of reporting all results (On-Demand DTCs) generated during the most recent diagnostic test executed.	6.2.8	Mandatory
75	Before an ECU shall be permitted to execute a Diagnostic Command, the ECU must be in the Diagnostic State.	7.4	Optional
76	Upon completing the execution of a Diagnostic Command, an ECU shall automatically return to the Diagnostic State.	7.5.1	Optional
77	During execution of a Diagnostic Command, an ECU shall terminate the Diagnostic Command and return to the Operational State if the ECU receives a Request Return to Operational State message.	7.5.2	Optional
78	During execution of a Diagnostic Command, an ECU shall terminate the Diagnostic Command and return to the Operational State if the ECU does not receive a valid node to node message within a 5.0 second (-0.0, +1.0) interval.	7.5.2	Optional
79	During execution of a Diagnostic Command, an ECU shall terminate the Diagnostic Command and return to the Operational State if the ECU satisfies any of the exit criteria specified in the ECUs SSDS.	7.5.2	Optional
80	Upon receiving the first Latching PID Command, an ECU shall clear (i.e., zero out) PID \$0201.	7.7.1	Optional
81	During a Latching PID command, if the first transition is low to high, the bit value latched into the PID shall be high - "1".	7.7.1	Optional

Requirement		Section	Mandatory or Optional
82	During a Latching PID command, if the first transition is high to low, the bit value latched into the PID shall be low - "0".	7.7.1	Optional
83	After an ECU receives the first Latching PID command, the ECU shall latch all initial I/O transitions and increment PID \$0201 each time an I/O transition is sensed.	7.7.1.1	Optional
84	After an ECU receives a second Latching PID command, the ECU shall stop monitoring its I/O lines and return the PID associated with those I/O lines to their normal state.	7.7.1.2	Optional
85	During a Latching PID command, an ECU shall stop monitoring its I/O lines and return the PID associated with those I/O lines to their normal state, if any of the Diagnostic Command entry criteria is violated.	7.7.1.2	Optional
86	An ECU shall be capable of returning PID information while executing in the Operational State.	8.2	Mandatory
87	An ECU shall be capable of returning PID information while executing in the No Stored Codes Logging Mode.	8.2	Optional
88	An ECU shall be capable of returning PID information while executing in the Diagnostic State.	8.2	Mandatory
89	An ECU shall be capable of returning PID information while executing in the Diagnostic Test Mode.	8.2	Mandatory
90	An ECU shall be capable of returning PID information while executing in the Diagnostic Command Mode.	8.2	Optional
91	An ECU shall be capable of returning PID information while executing in the Memory Transfer Mode.	8.2	Optional
92	A Tester shall retrieve PID information from an ECU through a Request Parameter by PID message.	8.6	Mandatory
93	Upon receiving a Request Parameter by PID message, an ECU shall return the requested PID data via the Report Parameter by PID message.	8.6	Mandatory
94	ECU I/O signal information returned in a PID shall indicate the current status of the ECU signals.	8.6	Mandatory
95	ECU I/O signal information returned in a PID shall not be subject to variability due to fault retry strategies.	8.6	Mandatory
96	An ECU shall support PID \$0200 - Number of continuous DTCs.	8.7	Mandatory
97	An ECU shall support PID \$0202 - Number of DTCs from most recent test.	8.7	Mandatory
98	An ECU shall support PID \$D100 - ECU Operating State / Mode.	8.7	Mandatory
99	An ECU shall support PID \$E200 - Software Version Number.	8.7	Mandatory
100	An ECU shall support PID \$E217 - Part Number Identification Base.	8.7	Mandatory
101	An ECU shall support PID \$E219 - Part Number Identification Suffix.	8.7	Mandatory
102	An ECU shall support PID \$E21A - Part Number Prefix.	8.7	Mandatory
103	An ECU shall support Continuous DTCs	9.2	Mandatory
104	An ECU shall support On-Demand DTCs	9.2	Mandatory
105	Each Continuous DTC logged by an ECU shall be maintained for a time period ranging from no less than 80 to no more than 128 ECU power up cycles.	9.2.1.1	Mandatory
106	When an ECU detects a fault and logs a Continuous DTC, it shall log a count in the "clearing counter" in conjunction with the fault.	9.2.1.2	Mandatory
107	Each time an ECU powers-up, the clearing counters for all logged Continuous DTCs shall be updated.	9.2.1.2	Mandatory

Requirement		Section	Mandatory or Optional
108	When an ECU detects a fault that already has a Continuous DTC logged, the clearing counter for that Continuous DTC shall be reset to its initial value.	9.2.1.2	Mandatory
109	When a Continuous DTC clearing counter's time period expires (between 80 and 128 power ups), the continuous DTC shall be cleared.	9.2.1.2	Mandatory
110	An ECU shall be capable of logging all Continuous DTCs and their associated clearing counters when the number of DTCs to be supported by an ECU is less than 15.	9.2.1.3	Mandatory
111	An ECU shall be capable of logging a minimum of 15 Continuous DTCs and their associated clearing counters when the number of DTCs to be supported by an ECU is 15 or more.	9.2.1.3	Mandatory
112	In response to a Request Parameter by PID message, with the address set to \$0200 - Number of Continuous DTCs, an ECU shall return a Report Parameter by PID message with the number of Continuous DTCs currently logged as the data.	9.2.1.4	Mandatory
113	In response to a single Request Stored Codes message, an ECU shall report all Continuous DTCs being logged by returning as many consecutive Report Stored Codes messages as is required.	9.2.1.4	Mandatory
114	An ECU shall support clearing of continuous DTCs manually.	9.2.1.5	Mandatory
115	An ECU shall support clearing of continuous DTCs automatically.	9.2.1.5	Mandatory
116	An ECU shall clear all Continuous DTCs it has logged upon receiving a Request Clear Stored Codes message; except DTC A141 and DTC A477.	9.2.1.5.1	Mandatory
117	Once all Continuous DTCs have been cleared, an ECU shall respond to the Request Clear Stored Codes messages with the General Response message of 00 - affirmative.	9.2.1.5.1	Mandatory
118	An ECU shall automatically clear a Continuous DTC logged when the fault which caused the continuous DTC to be logged is not detected for 80 to 128 power up cycles; except DTC A141 and DTC A477.	9.2.1.5.2	Mandatory
119	In response to a Request Parameter by PID message, address \$0202 -Number of Trouble Codes Set Due to Diagnostic Test, an ECU shall return a Report Parameter by PID message with the number of On-Demand DTCs generated during the most recent diagnostic test.	9.2.2.1	Mandatory
120	In response to a single Request Diagnostic Routine Results message, an ECU shall report all On-Demand DTCs being logged by returning as many consecutive Report Diagnostic Routine Results messages as is required.	9.2.2.1	Mandatory
121	On-Demand DTCs shall be cleared when the ECU returns to the Operational State or when the ECU executes another Diagnostic Test	9.2.2.3	Mandatory
122	An ECU shall not clear On-Demand DTCs upon receiving a Request Clear Stored Codes message.	9.2.2.3	Mandatory
123	An ECU shall support DTC \$9342 - ECU is Defective.	9.3	Mandatory
124	DTC \$9342 shall be logged by an ECU when it detects a failure with its internal circuitry while in the Operational State or while executing a test in the Diagnostic State.	9.3.1	Mandatory
125	An Invalid Data Network fault is a Continuous DTC that shall be logged by an ECU when "Invalid Data" is received in a SCP network message.	10.2.1	Mandatory

Requirement	Section	Mandatory or Optional
126 For digital invalid data, the transmitting ECU shall shift all data one byte position to the right, after the message's source address (byte 3), and place the digital invalid data value, \$3F, into the byte immediately following the source address.	10.2.1.1	Mandatory
127 If an ECU is responsible for providing analog data in a network message, detects a fault that would render all or part of that analog data invalid, the ECU shall replace each affected byte of normally valid data with the invalid data value of \$FF.	10.2.1.2	Mandatory
128 If an ECU that is responsible for providing digital and analog data in a network message, detects a fault that would render all or a portion of that data invalid, it shall follow the requirements defined for digital/analog invalid data.	10.2.1.3	Mandatory
129 The Missing Message Network Continuous DTC, \$D262, shall only be supported by an ECU if the missing message is deemed a critical failure to the function of the ECU.	10.3	Optional
130 If an ECU which supports the Missing Message Network DTC, \$D262, does not receive an expected message, the ECU shall set a missing message flag in RAM and initialize a 5.0 second (-0.0, +1.0) timer.	10.3.1	Optional
131 If an ECU which supports the Missing Message Network DTC, \$D262, receives the expected message during the 5.0 second interval, the ECU shall reset the unique missing message flag.	10.3.1	Optional
132 If an ECU supports the Missing Message Network DTC, \$D262, the ECU shall NOT log DTC \$D262 if all missing message flags are reset prior to the 5.0 second (-0.0, +1.0) timer expiration.	10.3.1	Optional
133 If an ECU supports the Missing Message Network DTC, \$D262, the ECU shall log DTC \$D262 in non-volatile memory if any missing message flags remain set after the 5.0 second (-0.0, +1.0) timer expires.	10.3.1	Optional
134 If a Display Driven ECU, which supports the "Missing Message" Network DTC \$D262, does not receive an expected message, the Display Driven ECU shall set a missing message flag in RAM and initialize/monitor a missing message timer.	10.3.2	Optional
135 The Display Driven ECU missing message timer shall only be initialized/reset when there are NO other missing message flags.	10.3.2	Optional
136 The Display Driven ECU missing message timers shall be either 1.0 second (-0.0, +0.25), 2.0 seconds (-0.0, +0.5), or 5.0 seconds (-0.0, +1.0).	10.3.2	Optional
137 If after the missing message time period has expired and the Display Driven ECU has not received the expected message, the Display Driven ECU shall log DTC \$D262	10.3.2	Optional
138 If the expected missing message is received before the missing message time period has expired, the Display Driven ECU shall reset the missing message flag.	10.3.2	Optional
139 If NO other missing message flags are set, the Display Driven ECU shall reset the missing message timer.	10.3.2	Optional
140 If all missing message flags are reset/cleared, DTC \$D262 shall NOT be logged by the Display Driven ECU.	10.3.2	Optional
141 A Display Driven ECU shall log DTC \$D262, whenever an event that is visible to the customer is caused because of a missing message.	10.3.2	Optional
142 An ECU shall NOT illuminate any visual indicator tied to the detection of a missing message fault until DTC \$D262 is logged in non volatile memory.	10.3.2	Optional
143 If an ECU transmits a Function Read Request message and receives an acknowledge that contains invalid data, the ECU shall log the associated invalid data network DTC into non volatile memory.	10.4.1.1	Mandatory

	Requirement	Section	Mandatory or Optional
144	If an ECU transmits a Function Read Request message and does not receive an acknowledge, the ECU shall NOT log a DTC.	10.4.1.1	Mandatory
145	If an ECU transmits a Status Request message and receives a response that contains invalid data, the ECU shall log the associated invalid data network DTC into non volatile memory.	10.4.2.1	Mandatory
146	If an ECU which supports the missing message network DTC, \$D262, transmits a Status Request message and does not receive a response, the ECU shall age a network DTC for a period of 5.0 seconds (-0.0, +1.0) before logging DTC \$D262 into non volatile memory.	10.4.2.1	Optional
147	For Event Driven Broadcast messages, upon each occurrence where a fault is present, the ECU transmitting the Event Driven Broadcast message shall send the message with invalid data.	10.4.3.1	Mandatory
148	For Event Driven Broadcast messages, when the fault is no longer present, the ECU shall return to sending valid data upon the occurrence of subsequent events.	10.4.3.1	Mandatory
149	If an ECU transmits an Event Driven Broadcast message and does not receive an acknowledge, the ECU shall NOT log a DTC.	10.4.3.1	Mandatory
150	If an ECU receives an Event Driven Broadcast message that contains invalid data, the ECU shall log an invalid data network DTC associated with that function into non volatile memory.	10.4.3.2	Mandatory
151	An ECU which receives Event Driven Broadcast messages shall NOT log a missing message network DTC, \$D262.	10.4.3.2	Optional
152	If an ECU which transmits a Periodic Broadcast message, detects an invalid data fault, the ECU shall transmit the message with invalid data for as long as the fault exists.	10.4.4.1	Mandatory
153	For Periodic Broadcast messages, if the invalid data fault is no longer present, the ECU shall send valid data.	10.4.4.1	Mandatory
154	If an ECU transmits a Periodic Broadcast message and does not receive an acknowledge, the ECU shall NOT log a DTC.	10.4.4.1	Mandatory
155	If an ECU receives a Periodic Broadcast message that contains invalid data, the ECU shall log an invalid data network DTC associated with the fault into non volatile memory.	10.4.4.2	Mandatory
156	If an ECU which supports the missing message network DTC, \$D262, does not receive a Periodic Broadcast message, the ECU shall age a network DTC for a period of 5.0 seconds (-0.0, +1.0) before logging DTC \$D262 into non volatile memory.	10.4.4.2	Optional
157	An ECU shall implement either a non sleep node network diagnostic strategy or a sleep node network diagnostic strategy, based on its node type.	10.4.5	Mandatory
158	A non sleep node ECU shall NOT start its diagnostic network fault strategy until 5.0 seconds (-0.0, +1.0) after its ignition run input, or equivalent signal input, has gone active.	10.4.5.1	Mandatory
159	A Sleep Node ECU shall NOT start its diagnostic network fault strategy until 5.0 seconds (-0.0, +1.0) after it has received an Ignition Switch Position - RUN message, or equivalent message, over the SCP serial communications link.	10.4.5.2	Mandatory
160	A sleep node ECU shall discontinue its Diagnostic Network Fault Strategy upon receiving a message which states that the vehicle's Ignition switch is no longer in the RUN position.	10.4.5.3	Mandatory
161	An ECU that supports logging of a Missing Message Network DTC shall implement the No Stored Codes Logging Mode.	10.5	Optional
164	While in the No Stored Codes Logging mode, an ECU shall NOT log ANY Continuous DTCs.	10.5.1	Mandatory
163	While in the No Stored Codes Logging mode, an ECU shall provide access to parameter Identifier information.	10.5.1	Mandatory

Requirement		Section	Mandatory or Optional
164	While in the No Stored Codes Logging mode, an ECU shall provide data access via direct memory reference (DMR).	10.5.1	Optional
165	While in the No Stored Codes Logging mode, an ECU shall provide entry into the Diagnostic State.	10.5.1	Mandatory
166	While in the No Stored Codes Logging mode, an ECU shall respond normally to all messages received from a Tester.	10.5.1	Mandatory
167	While in the No Stored Codes Logging mode, an ECU shall NOT respond with "Invalid Data" to messages received from other ECUs on SCP.	10.5.1	Mandatory
168	While in the No Stored Codes Logging mode, an ECU shall respond with "Valid Data" to all messages received from other ECUs on SCP.	10.5.1	Mandatory
169	Upon receiving a Request No Stored Codes Logging message while in the Operational State, an ECU shall enter the No Stored Codes Logging mode.	10.5.1.1	Mandatory
170	Upon receiving a Request No Stored Codes Logging message while in the Operational State, an ECU shall initialize and enable a 5.0 second (-0.0, +1.0) "Remain in the No Stored Codes Logging Mode" timer.	10.5.1.1	Mandatory
171	Upon receiving a Request No Stored Codes Logging message while in the Operational State, an ECU shall update PID \$D100 - ECU Operating State / Mode to \$09, No Stored Codes Logging Mode.	10.5.1.1	Mandatory
172	An ECU shall exit the No Stored Codes Logging mode and return to the Operational State the ECU receives a Request Return to Operational State message.	10.5.1.2	Mandatory
173	An ECU shall exit the No Stored Codes Logging mode and return to the Operational State the ECU does not continue to receive the functionally broadcast Request No Stored Codes Logging message within every 5.0 seconds (-0.0, +1.0).	10.5.1.2	Mandatory
174	Upon returning to the Operational State, from the NSCL mode, an ECU shall return to normal operation (i.e., continue logging Continuous DTCs) within 500 milliseconds.	10.5.1.3	Mandatory
175	An ECU shall exit the No Stored Codes Logging mode and enter the Diagnostic State if the ECU receives a Request Diagnostic State Entry message.	10.5.1.4	Mandatory
176	When a Request Parameter by DMR message is received by the ECU which contains the address of a sensitive memory location, or out of range data, or an invalid address; the ECU shall report a General Response message, response code of \$12; Invalid Data.	11.2.1	Optional
177	The Security Access Procedure shall be used when downloading data to an ECU and intentional modification of the data could cause harm to the occupants of a vehicle or to the subsystem.	11.3	Optional
178	While in the Operational State, Diagnostic State, Test Mode, Command Mode, or Memory Transfer Mode, the gateway module shall respond directly to all messages directed at it.	12.2	Optional
179	During initialization, after power up or reset, the gateway module shall evaluate the interface to each of the satellite modules.	12.2.1	Optional
180	If a gateway module can not communicate with one or more of the satellite modules during initialization, the gateway module shall log a DTC to identify this fault condition.	12.2.1	Optional
181	During execution of the on demand self test, the gateway module shall evaluate the interface to each satellite module.	12.2.2	Optional

Requirement		Section	Mandatory or Optional
182	If a gateway module can not communicate with one or more of the satellite modules during the execution of the on demand self test, the gateway module shall log a DTC to identify this fault condition.	12.2.2	Optional
183	To enter the gateway mode, the gateway module shall be in the diagnostic state.	12.3.1	Optional
184	To enter the gateway mode, the gateway module shall receive diagnostic command \$0300 - gateway mode access.	12.3.1	Optional
185	Upon receipt of the diagnostic command \$0300 - gateway mode access, the gateway module shall set up a pointer to the satellite module specified in byte 7 of the diagnostic command \$0300 message.	12.3.1	Optional
186	While in the gateway mode, the gateway module shall translate messages received from a tester into messages the satellite modules support and route the message to the correct satellite module.	12.3.1.1	Optional
187	While in the gateway mode, the gateway module shall be capable of translating all diagnostic messages supported by the satellite modules.	12.3.1.1	Optional
188	While in the gateway mode, the gateway module shall translate messages received from a satellite module into a protocol that it supports and route the message to the tester.	12.3.1.1	Optional
189	The gateway module shall support the diagnostic command \$FACE - Hold Gateway Mode.	12.3.1.2	Optional
190	All message received by the gateway module from a tester that are formatted correctly shall be translated into a protocol supported by the satellite modules and routed to the satellite module identified in byte 7 of the diagnostic command.	12.3.1.3	Optional
191	If the gateway module receives diagnostic command \$0300 - gateway mode access, it shall return a General Response message with the response code set to \$00 - affirmative.	12.3.1.3	Optional
192	If the gateway module receives diagnostic command \$0300 - gateway mode access, with byte 8 = \$10, it shall return a General Response message with the response code set to \$00 - affirmative, and exit the gateway mode to the diagnostic state.	12.3.1.3	Optional
193	If the gateway module receives diagnostic command \$0300 - gateway mode access, with byte 8 = \$20, it shall return a General Response message with the response code set to \$00 - affirmative, and exit the gateway mode to the operational state.	12.3.1.3	Optional
194	Upon returning to the operational state, the gateway module shall have 500 milliseconds to reinitialize, during which time the gateway module is not required to respond to any messages received from the tester.	12.3.1.3	Optional
195	If the gateway module receives diagnostic command \$0300 - gateway mode access, with byte 8 not equal to \$00, \$10, or \$20, it shall return a General Response message with the response code set to \$12 - invalid data .	12.3.1.3	Optional
196	If the gateway module receives diagnostic command \$0300 - gateway mode access, with byte 7 containing a value that is not a recognized satellite module, it shall return a General Response message with the response code set to \$12 - invalid data .	12.3.1.3	Optional
197	If the gateway module receives diagnostic command \$FACE - Hold Gateway Mode, it shall return a General Response message with the response code set to \$00 - affirmative.	12.3.1.3	Optional
198	If the gateway module receives diagnostic command \$FACE - Hold Gateway Mode, it shall inhibit the time out function that causes it to return to operational state if it does not receive a valid node to node message within 5.0 seconds.	12.3.1.3	Optional

Requirement		Section	Mandatory or Optional
199	Upon receiving a diagnostic command \$FACE-Hold Gateway Mode, a second time, the gateway module shall re-establish the time out function so that it may return to the Operational State if it does not receive a valid node to node message within 5.0 seconds.	12.3.1.3	Optional
200	If a Gateway module receives diagnostic command \$FACE - Hold Gateway Mode, while not in the gateway mode, the Gateway module shall respond with a General Response message with the response code set to \$22 - Conditions Not Correct.	12.3.1.3	Optional
201	If the Gateway module receives a diagnostic message that is not supported, it shall respond with a General Response message with the response code set to \$11 - Mode Not Supported.	12.3.1.3	Optional
202	If a Satellite module receives a translated message from the Gateway Module that is supported and formatted correctly, it shall formulate a proper response and return it to the Gateway module.	12.3.1.3	Optional
203	If a Satellite module receives a message from the Gateway Module that is not supported, it shall return a message that can be translated by the Gateway module into General Response message with the response code set to \$11 - Mode Not Supported.	12.3.1.3	Optional
204	If a Satellite module receives a message from the Gateway Module that contains invalid data, it shall return a message that can be translated by the Gateway module into General Response message with the response code set to \$12 - Invalid Data.	12.3.1.3	Optional
205	Any messages received from Satellite modules that are not currently pointed to by the Gateway module shall be ignored by the Gateway module.	12.3.1.3	Optional
206	Any messages received from Satellite modules that do not correspond to the request sent, shall be ignored by the Gateway module.	12.3.1.3	Optional
207	Any messages received from Satellite modules that are not formatted correctly, shall be ignored by the Gateway module.	12.3.1.3	Optional
208	The Gateway module shall exit the gateway mode and return to the Diagnostic State if it receives the Diagnostic Command \$0300 - Gateway Mode Access, with byte 8 set to \$10 - Return to Diagnostic State.	12.3.2	Optional
209	The Gateway module shall exit the gateway mode and return to the Operational State if it fails to receive a valid node to node message within 5.0 seconds (-0.0, +1.0).	12.3.3	Optional
210	The Gateway module shall exit the gateway mode and return to the Operational State if it receives the Diagnostic Command \$0300 - Gateway Mode Access, with byte 8 set to \$20 - Return to Operational State.	12.3.3	Optional
211	The response time between the trailing edge of the last bit of the request message issued by the tester and the leading edge of the response message returned to the tester shall not exceed 275 milliseconds.	12.4	Optional
212	The time required for a gateway module to translate from one protocol to another (tester to satellite module/satellite module to tester), shall not exceed 20 milliseconds.	12.4	Optional
213	A satellite module shall remain in the diagnostic state as long as it receives a valid node to node message with 5.0 second (-0.0, +1.0) intervals or until it is instructed to exit.	12.5.1	Optional
214	All Satellite modules shall support PID \$D12D - Node Address.	12.5.2	Optional

Appendix C CHANGE HISTORY

No	Sect.	Change(s):	Reason for change
1	8.6.3	\$03 - \$07 : should say 'state' according to MRDB \$08 : in MRDB, not in specification \$09 : conflicting descriptions between spec. and MRDB \$0A - \$0C : in MRDB, not in specification and \$85 : should say 'state' according to MRDB	States in PID \$D100 conflict with the MRDB.
2	8.6.5	"part number PIDs must be read from byte 1 to byte 4" the change here overwrites the verbiage in the reason for change column.	The wording for section 8.6.5 is incorrect specifying "part # PIDs characters must begin with Byte 4 and progress through Byte 1". The part number is actually read in the opposite direction.
3	11.4.5	Add flowcharts and descriptions explaining ECU upload procedures. This is a newly added section	There aren't any flowcharts or descriptions of the diagnostic upload memory transfer in the main document. There are only messages defined in the appendix.
4	A-24	Add to the responses for Memory Transfer mode to include general responses of 12 and 21 for sub-function not supported and busy. Also, remove all references to uploading (I.E. ECU Message: that contains data to be uploaded from the ECU to the tester.	Message responses specified in flow 11-3 weren't reflected in the appendix for block data transfer message, mode 36. Also, the references to upload with this mode need to be removed since it will be replaced with a mode 75.
5	Fig. 3-2	Remove the return from command mode to operational state specified by a message \$B1 for return.	Correct error. This mode is to issue a command, not to exit command mode.
6	Fig. 4-1	Added "500 msec init. allowed" to the description text associated with the return to operational state from diagnostic state.	Clarification.
7	Fig. 4-1	Reversed the direction of the general response arrow coming from the ECU in operational state.	Correct error.
8	5	Deleted the text under the heading I/O in the Diagnostic State.	Removed here and added to another section for clarification.
9	5.3	Re-worded verbiage for section 5.3 to clarify the requirements of the module to stay in diagnostic state.	Clarification.
10	5.3	Added the following statements regarding diagnostic state requirements: 1) All ECU controlled outputs shall be made inactive, unless specified otherwise in the ECUs SSDS. 2) All ECU inputs shall be isolated from directly controlling related ECU outputs (e.g., switch input will not activate output).	Moved in document for clarification.

No	Sect.	Change(s):	Reason for change
11	Fig. 5-1	Reversed the direction of the arrows for modes \$32 and \$35. Also, "Report block" text replaced "Read memory block" for mode \$7C.	Error correction
12	5.4	Added "The ECU doesn't receive ANY Node-to-Node message (supported or unsupported) within 5.0 (-0.0, +1.0) seconds of either the last message received or initial entrance into diagnostic state."	Clarification
13	5.4	Eliminated "Auto-Clearing Diagnostic Test Results" section.	Is redundant and adequately described in later section
14	6.2.5.2	Added "The ECU doesn't receive ANY Node-to-Node message (supported or unsupported) within 5.0 (-0.0, +1.0) seconds of either the last message received or initial entrance into diagnostic state."	Clarification
15	6.2.8	"Clearing of" was removed in the section describing clearing of diagnostic test results and "shall be cleared" was added.	Clarification
16	7.5	Added "The ECU doesn't receive ANY Node-to-Node message (supported or unsupported) within 5.0 (-0.0, +1.0) seconds of either the last message received or initial entrance into diagnostic state."	Clarification
17	8.7.4	Added "ASCII" for a PID type.	Error correction. Just wasn't in spec.
18	10.5.1.2	Deleted the "continue to" text and added "within 5.0 (-0.0, +1.0) seconds of either the last broadcast message received or initial entrance into No Stored Codes Logging Mode.)"	Clarification
19	Fig. 10-6	Reversed the general response arrow direction.	Error correction
20	Fig. 10-6	Removed the SCP reference	Eliminating for future protocol additions
21	11.4	Removed all of the document number references and referred to Appendix D.	Easier maintenance and does reside in appendix
22	12.3.3	Added "The ECU doesn't receive ANY Node-to-Node message (supported or unsupported) within 5.0 (-0.0, +1.0) seconds of either the last message received or initial entrance into gateway mode and"	Clarification
23	A-10	Eliminated references to DTC number since this can be any multiple of 3. Also, changed the text from "will be reported" to "transferred for this response". For responses of mode \$13 "request continuous codes."	Clarification
24	A-22	Eliminated references to DTC number since this can be any multiple of 3. Also, clarified description text. For mode \$34 "request download routine entry" responses.	Clarification

No	Sect.	Change(s):	Reason for change
25	A-25	Replaced the description wording to "Affirmative for data received. Handshake for Format 80 responds to each block transferred. Non- handshake for at 81 responds after all blocks have been transferred." for affirmative response, added "Invalid data was present in the block transfer message" for sub-function not supported, and added module busy response. For mode \$36.	Addition and clarification
26	A-26	Removed the affirmative responses for operational state and no stored codes logging modes. Also, added mode not supported for test mode, command mode and memory transfer mode. For mode \$3B	Not safe or necessary to be writing to memory locations while in operational state or no stored codes logging mode. Also, needed mode not supported for when the module doesn't support write memory blocks.
27	A-27	Same as change number 26 for page A-26 above.	Same as change number 26 for page A-26 above.
28	Appx. D	Added reference to "Module Configuration Specification" and its part #	Document maintenance
29	Tab. 8-7	Added "*" = indicates no alternate"	Indicates that there aren't any alternate parts for these encodings
30	2.4	Changed the timing of FMEA completion from Program Readiness to Program Approval	Error correction. The FPDS timing states that FMEAs are to be completed by Program Approval
31	A-18	"Security algorithm data byte 'X' for request seed response when byte 5 = 1, \$00 otherwise" was added for response mode \$67 for bytes 9 and 10 of the response message.	The response was incorrect specifying that the byte value was always zero
32	9.5	Add section in chapter 9 explaining when and how to use circuit failure/ malfunction DTCs. Section 9.5 was added specifying to use DTCs with the description of malfunction/failure circuit when the nature of the circuit fault cannot be determined.	Addition to specify when and how to use circuit failure/malfunction DTCs
33	11.4	Eliminated all references to "Echo"	Clarification. There isn't a difference between echo and handshake. Echo was eliminated to simplify the description of the upload and download procedures
34	Fig 11-6	Added new flowchart for upload without handshake.	New part 1 content. Modules starting to use functionality and specification didn't define method
35	Entire	Made numerous grammatical changes.	Correction
36	6.2.4	Added section describing tester to ECU downloaded	Pulled in from SCP - 002 document,

No	Sect.	Change(s):	Reason for change
		execution routines.	wanted to allow ability
37	Appx. D.3	Added references to J1213, J1978, and J2186	Pulled in from SCP - 002 document
38	Appx. E	New appendix containing definitions relating to multiplex and diagnostics	Pulled in from SCP - 002 document
39	3.2.2.2	Added references to SCP 001 and SCP 003. Also, removed reference to SCP - 002 document reference.	Pulled in from SCP - 002 document
40	3.2.2.3	Added ability for co-existence of SCP-CARB and SCP-DIAG on the same network.	Pulled in from SCP - 002 document
41	A.3 and A.3.1 new	Added a general heading for SCP Diagnostic Description and a sub-heading for qualifying SCP message responses from a tester.	Pulled in from SCP - 002 document
42	A.3.2 new	Added section describing SCP-CARB diagnostic capabilities.	Pulled in from SCP - 002 document
43	A.2.1	Eliminated wording stating that all diagnostic messages are physically addressed except for the No Stored Codes Logging Mode request message. Put a reference to the No Stored Codes Logging Mode request as an example instead.	SCP-CARB is functional based and allowed. Needed to eliminate the restrictions on functional addressing that would restrict using SCP - CARB
44	D.2	Modified part number suffixes to reflect the latest revisions on the reference documents listed.	Maintenance/accuracy.
45	3.2.2.2	Added reference to section 5.3 of SCP-003 referring to maximum length of off-board tester bus extension cable.	Pulled in from SCP - 002 document
46	3.2.2.2	Added interface circuitry reference to section 5.6 of SCP-003.	Pulled in from SCP - 002 document
47	8.4 new	Added section "PID SCALING AND MASKING CAPABILITY".	Needed description for new modes \$24 and \$64 added for PID scaling and masking capability. Powertrain first users of this capability. Mode definitions/descriptions taken from MUX - 002 documents
48	A.17 and A.18 new	Added message tables to the appendix for modes \$24 and \$64	Used to define the message bytes for the newly added modes of \$24 and \$64
49	3.2.2.1	Added reference to the ISO MUX - 002 document for the tester and ECU physical layers.	User added reference
50	A.3.6, A.3.6.1 & A.3.6.2 All new	Added all relevant information from ISO - 002 document relating to message gaps.	Pulled in from ISO - 002
51	A.3.3, A.3.3.1 & A.3.3.2	Added all relevant information from SCP - 002 document relating to message gaps.	Pulled in from SCP - 002

<u>No</u>	<u>Sect.</u>	<u>Change(s):</u>	<u>Reason for change</u>
52	A.3	Added ISO information relating to byte bit encoding	Pulled in from ISO - 002
53	A29	Added description to number of bytes to be downloaded to include checksum bytes.	Clarification
54	7.1	Made data size for all commands in Table 7-1 "1-4 bytes"	Error correction